PhysSANet: Formation-Inspired Perturbations and Structure-Aware Learning for Semi-Supervised Remote Sensing Change Detection

Lin Chen, Zimei Wei, Shukan Liu, Qiao Sun, Xianfeng Zhang, and Bin Hu

***Abstract*—Remote sensing change detection (CD) is constrained by the high cost of pixel-level annotations, which limits the scalability of deep models for large-area monitoring. Semi-supervised learning (SSL) can leverage unlabeled data, yet existing approaches often rely on generic natural-image augmentations and structure-agnostic representations, resulting in unreliable pseudo-labels under remote-sensing-specific appearance variations such as haze-like veiling, illumination/exposure differences, and sensor noise. In this article, we propose a physics-inspired and structure-aware network (PhysSANet), a physics-inspired and structure-aware SSL framework for remote sensing CD. PhysSANet introduces a physics-aware perturbation (PAP) strategy that simulates common acquisition-related variations by sampling parameters from typical ranges and adapting them using image statistics. PAP is metadata-free and does not attempt to recover physical quantities. To further improve boundary-accurate localization, PhysSANet couples PAP with a multihead structure-aware attention (MHSAA) module and a spatial–structural attention-enhanced feature pyramid network (SSA-FPN), which explicitly model geometric cues including boundaries, regional coherence, and multiscale spatial structure. Moreover, we design a multi-view consistency regularization (MVCR) scheme that integrates teacher–student consistency with appearance-plausibility-aware pseudo-label filtering and structure-aware smoothness, suppressing unreliable pseudo-labels under strong appearance variations. Extensive experiments on three benchmark datasets (LEVIR-CD, WHU-CD, and SYSU-CD) demonstrate the effectiveness of PhysSANet across varying supervision levels. With only 5% labeled data, PhysSANet achieves 82.3% intersection over union (IoU) on WHU-CD, 81.9% on LEVIR-CD, and 63.6% on SYSU-CD, consistently outperforming strong semi-supervised baselines. These results demonstrate that combining physics-inspired perturbations with structure-aware representation learning improves label efficiency, robustness, and boundary accuracy for semi-supervised remote sensing CD.**

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I. Introduction

CHANGE detection (CD) in remote sensing aims to identify land-cover changes from multitemporal observations and supports a wide range of applications such as urban monitoring, disaster response, and environmental management [1], [2], [3]. With the rapid growth of high-resolution satellite imagery featuring improved spatial, spectral, and temporal characteristics [4], [5], [6], CD models are increasingly expected to deliver fine-grained and timely change maps over large areas. However, imagery acquired across different times, regions, and sensors often exhibits substantial acquisition-induced inconsistencies [7], [8], [9], [10], including atmospheric veiling (e.g., haze-like effects), illumination/exposure shifts, and sensor/viewpoint-related variations. These appearance and geometry discrepancies can markedly degrade CD generalization, especially when deployment conditions deviate from the training domain.

Meanwhile, collecting high-quality pixel-level change annotations at scale remains prohibitively expensive for long-term and large-area remote sensing monitoring [1], [11], [12]. Annotating a single city-scale bitemporal dataset typically requires hundreds of person-hours and substantial domain expertise, which severely limits the practicality of fully supervised learning in real-world operational settings. As a result, a pronounced gap arises between practical deployment requirements and the capabilities of current deep-learning-based CD methods, which still heavily depend on abundant pixel-level annotations [13], [14]. This challenge motivates the adoption of semi-supervised learning (SSL) for CD, where large amounts of unlabeled imagery can be effectively exploited to alleviate annotation costs while maintaining competitive performance [15], [16], [17].

Despite recent progress, existing SSL-based CD methods face two critical limitations in real-world remote sensing scenarios and large-scale deployment settings. First, most consistency-based SSL frameworks adopt perturbations designed for natural images (e.g., color jittering and random cropping) [18], [19]. Such perturbations are often mismatched to common acquisition-related variations in remote sensing and may introduce implausible appearance changes that distort

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scene statistics or weaken the correspondence of bitemporal pairs. Consequently, consistency supervision may become less informative under severe acquisition-induced variations, leading to noisy pseudo-labels and unstable training. Second, accurate CD requires precise delineation of object boundaries and coherent spatial structures, especially in complex urban scenes [20], [21]. However, many SSL frameworks enforce consistency primarily at the feature or prediction level without explicitly leveraging structural priors; the resulting pseudo-labels, therefore, tend to suffer from boundary noise, broken thin structures, or spatial fragmentation. Although methods such as C2F-SemiCD [22] employ coarse-to-fine refinement, a systematic coupling between acquisition-aligned perturbations and structure-aware regularization remains underexplored.

To bridge this gap, we argue that SSL for remote sensing CD should jointly consider: 1) *acquisition-aligned appearance perturbations* to make consistency learning realistic and robust and 2) *explicit structural modeling* to preserve boundary precision and spatial coherence. Importantly, such a design should improve robustness without requiring per-image metadata or performing physical inversion, which is often unavailable or impractical in large-scale scenarios.

Motivated by these insights, we propose a physics-inspired and structure-aware network (*PhysSANet*), a semi-supervised CD framework that unifies acquisition-aligned perturbations and geometric priors. Here, "physics-inspired" is used in a lightweight and pragmatic sense: it denotes formation-inspired, parameterized perturbations applied in the observable image domain to mimic typical acquisition-induced variations (e.g., haze-like veiling, illumination/exposure shifts, and sensor noise), instead of physically exact forward simulation, atmospheric correction, or inverse recovery of physical variables. Therefore, PhysSANet is metadata-free and noninversion by design, and it does not assume access to per-image acquisition geometry or atmospheric parameters. Moreover, different from physics-informed neural network (PINN)-style approaches that impose partial differential equation (PDE) residuals or explicit physical constraints, we inject only weak acquisition priors into the SSL pipeline via controllable perturbation families and observable-domain plausibility checks, aiming to improve pseudo-label reliability and boundary accuracy under realistic remote sensing appearance variability.

PhysSANet consists of four tightly coupled components.

1) *A physics-aware perturbation* (PAP) module that parameterizes and constrains typical acquisition-related variations (atmospheric veiling, illumination/exposure changes, and sensor noise) to generate weak/strong training views without relying on metadata or recovering physical quantities. PAP is designed for robust SSL under acquisition variability and is not intended for atmospheric correction, radiometric calibration, or any physical-quantity estimation.
2) *A multihead structure-aware attention* (MHSAA) module that captures boundary-level and region-level geometric cues across time.
3) *A spatial–structural attention-enhanced feature pyramid network* (SSA-FPN) that preserves fine structures during multiscale feature fusion.
4) *A multiview consistency regularization* (MVCR) scheme that integrates teacher–student consistency with appearance-plausibility-aware pseudo-label filtering and structure-aware smoothness to improve pseudo-label reliability under strong acquisition-induced variations.

By construction, PAP encourages invariance to realistic acquisition variability, while the structure-aware modules and MVCR guide the model toward spatially coherent and boundary-accurate change localization.

The main contributions are summarized as follows.

1) We propose PhysSANet, a physics-inspired and structure-aware semi-supervised framework for remote sensing CD that improves label efficiency and robustness under common acquisition-induced appearance variations, without requiring per-image metadata or explicit physical inversion.
2) We introduce PAP, an acquisition-aligned perturbation strategy that produces weak/strong views via parameter-range sampling and statistics constraints to emulate atmospheric veiling, illumination/exposure shifts, and sensor noise, thereby strengthening consistency learning and pseudo-labeling under appearance variability.
3) We design MHSAA and SSA-FPN to model multiscale structures and boundaries, enhancing change localization and boundary delineation in complex scenes.
4) We develop MVCR, which combines teacher–student consistency with appearance-plausibility-aware pseudo-label filtering and structure-aware smoothness to suppress unreliable pseudo-labels under strong perturbations; extensive experiments on three benchmarks validate consistent gains over strong semi-supervised baselines under low annotation ratios.

II. Related Work

We review three research directions most relevant to this study, including deep learning for remote sensing CD, SSL for dense prediction and CD, and physics-informed/physics-guided machine learning in remote sensing. Then, summarize the key gaps that motivate PhysSANet.

A. Deep Learning for Remote Sensing CD

Deep learning has become the dominant paradigm for remote sensing CD, evolving from handcrafted differencing to end-to-end representation learning. Siamese convolutional neural network (CNN)-based models, such as FC-EF [23] and SNUNet [24], exploit weight sharing and multiscale feature fusion to capture spatial structures, while Transformer-based designs (e.g., BIT [25] and ChangeFormer [26]) further introduce long-range context modeling. These frameworks achieve strong performance under full supervision, yet practical deployment often exposes two limitations.

First, *heavy annotation dependence*: most deep CD models require a large amount of pixelwise annotations to reach stable performance, and recent surveys report that many published methods are evaluated on datasets with thousands of labeled pairs [1], [3]. This limits scalability in large-area and long-term monitoring, where exhaustive labeling is costly and often

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TABLE I

COMPARISON OF REPRESENTATIVE SSL METHODS USED FOR OR ADAPTED TO CD. WE COMPARE AUGMENTATION/PERTURBATION STRATEGIES, CONSISTENCY MECHANISMS, AND WHETHER THE DESIGN IS ACQUISITION-ALIGNED AND STRUCTURE-AWARE

Method	Augmentation / Perturbation	Consistency Mechanism	Acq.-Aligned	Structure-Aware	Multi-Scale Fusion
General SSL (adapted for CD)					
Mean Teacher [27]	Flip+Crop	EMA Teacher	✗	✗	✗
FixMatch [28]	RandAugment	Weak–Strong	✗	✗	✗
UniMatch [29]	RandAugment	Dual-Stream	✗	✗	✗
Remote sensing-specific SSL					
SemiCD [30]	Color Jitter	Cross-Consistency	✗	✗	✗
S4GAN [31]	Adversarial	GAN-based	✗	✗	✗
SemiCDNet [32]	RandAugment	Confidence Threshold	✗	✗	✓
C2F-SemiCD [22]	RandAugment	Multi-view + masked	✗	Partial*	✓
ISCDNet [19]	Channel-wise	Context-Aware	✗	Data-Driven	✓
PhysSANet (Ours)	**Physics-Aware**	**Multi-view + masked**	✓	✓	✓

*Employs multi-scale attention, but does not impose explicit geometric constraints in the SSL objective.

infeasible. Second, *sensitivity to acquisition variability*: many CD networks are trained under relatively constrained imaging conditions and do not explicitly account for acquisition-induced appearance variations (e.g., atmospheric veiling, illumination/exposure shifts, and sensor/viewing geometry differences). Consequently, performance can degrade when the test domain exhibits different radiometric–geometric conditions since learned features may not be sufficiently invariant to such variations. These issues motivate the acquisition-aligned perturbation design (Section III-B) and structure-aware representation learning (Section III-C) in PhysSANet.

B. SSL for CD

SSL leverages unlabeled data via consistency regularization, pseudo-labeling, or contrastive learning [33]. Representative methods include Mean Teacher [27], which stabilizes pseudo-labels using an exponential moving average (EMA) teacher, FixMatch [28], which couples weak–strong perturbations with confidence thresholding, and CPS [34], which uses cross-pseudo supervision for dense prediction. These paradigms provide a widely adopted basis for SSL-based CD.

Recent works tailor SSL to CD with domain-specific designs (Table I). C2F-SemiCD [22] adopts a coarse-to-fine framework with hierarchical refinement, ISCDNet [19] improves pseudo-labels via context modeling, and SemiCDNet [32] applies confidence-based selection. While these methods improve pseudo-label quality, their perturbation strategies are still largely inherited from natural-image SSL and can be mismatched with remote sensing acquisition characteristics.

Specifically, widely used perturbations (color jitter, RandAugment, and random crops) can be problematic for bitemporal remote sensing imagery in at least three aspects.

1) *Radiometric Plausibility:* Unconstrained color/intensity transformations may push local statistics beyond plausible ranges for certain materials (e.g., water or shadow regions), producing unrealistic observations and weakening consistency targets.
2) *Radiometric–Geometric Coupling:* Real remote sensing appearance variations are often coupled with illumination/viewing geometry (e.g., solar angle and bidirectional reflectance distribution function (BRDF) effects), whereas generic perturbations ignore such coupling and may generate variations that poorly reflect operational conditions.
3) *Pairwise Correspondence:* Aggressive spatial transforms, especially when applied inconsistently across the bitemporal pair, can introduce misalignment-like artifacts that resemble changes and encourage learning spurious patterns.

These mismatches can reduce the reliability of consistency learning and pseudo-labeling under acquisition-induced variations. In contrast, PhysSANet introduces a PAP strategy (Section III-B) to generate acquisition-aligned weak/strong views, and stabilizes SSL training via structure-aware regularization (Section III-C) in a framework.

C. Physics-Informed/Physics-Guided Machine Learning in Remote Sensing

PINNs [35], [36] incorporate physical laws by penalizing PDE residuals while physics-aware learning has been explored for tasks such as remote sensing image dehazing [37] and environmental modeling [38]. More broadly, physics-based priors of image formation and environmental processes can improve data efficiency and robustness by constraining learning with domain knowledge.

However, classical PINNs are often computationally expensive and primarily target low-dimensional PDE systems, which are not directly suited to large-scale, high-resolution image understanding. In this work, we adopt a lightweight *physics-guided* perspective tailored to SSL-based CD: rather than enforcing PDE constraints or performing physical inversion, we inject acquisition-related priors into the learning pipeline in the observable image domain. Concretely, we design a PAP module that parameterizes typical acquisition-induced variations and constrains them using image statistics, enabling realistic perturbations even without per-image metadata. We further develop an MVCR scheme with plausibility-aware pseudo-label filtering and structure-aware smoothness, which suppresses unreliable pseudo-labels caused by severe appearance distortions.

Overall, although physics-inspired modeling has shown benefits in various remote sensing tasks, its integration with SSL-based CD remains limited. PhysSANet addresses this gap by coupling acquisition-aligned perturbations with structure-aware SSL regularization to improve robustness and label efficiency. In short, our "physics-inspired" design is a lightweight, observable-domain perturbation prior for SSL

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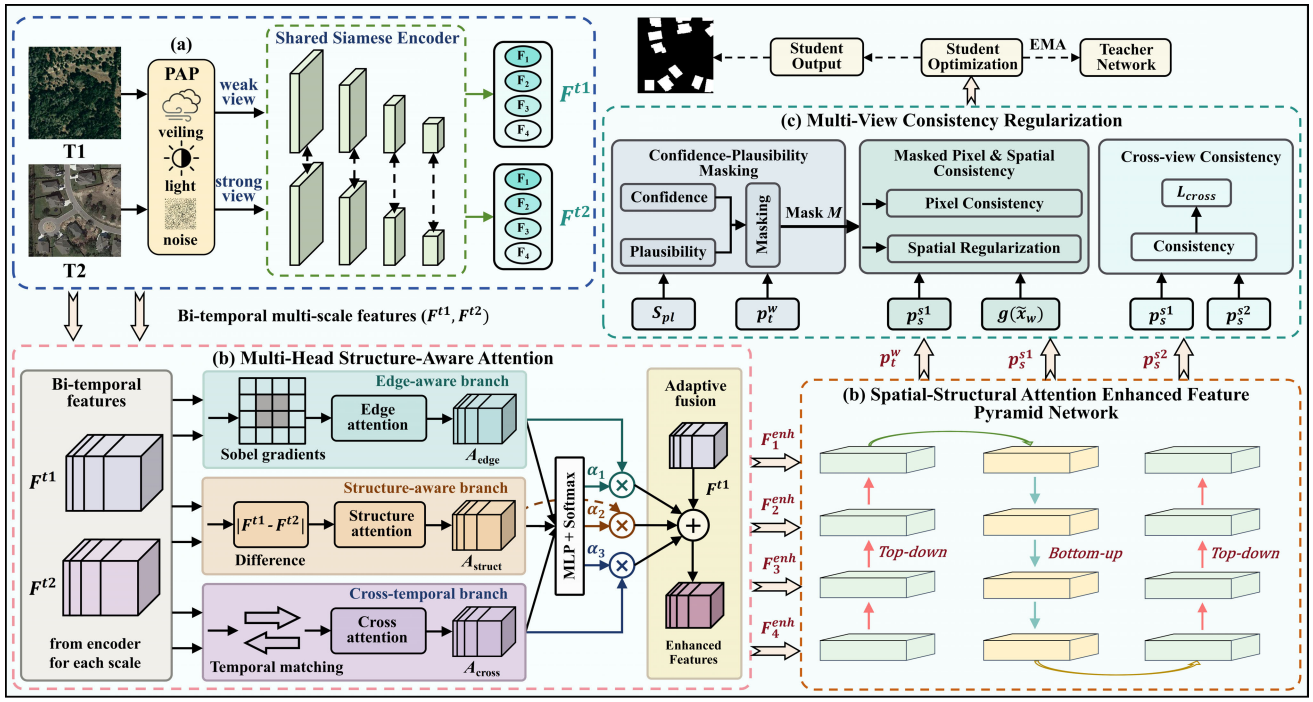


Fig. 1. Overview of the proposed PhysSANet framework. (a) PAP generates acquisition-aligned weak and strong views by simulating plausible appearance variations in remote sensing imagery, including atmospheric veiling (haze-like effects), illumination/exposure differences, and sensor noise. (b) Feature extraction and enhancement: bitemporal images are processed by a Siamese encoder, followed by the MHSAA module and the SSA-FPN to learn structure-preserving, multiscale representations that emphasize boundaries and region coherence. (c) MVCR: a teacher–student framework enforces consistency between predictions on weak and strong views. Teacher predictions on the weak view are filtered by a joint confidence and appearance-plausibility criterion to construct a reliability mask, which is then used to supervise the student on the strong view via masked unsupervised losses with structure-aware regularization.

robustness, rather than PINN-style physics constraints or any form of physical inversion.

D. Summary

In summary, prior SSL-CD methods insufficiently account for acquisition-induced variability and geometric structure. Generic perturbations can be poorly matched to realistic remote sensing appearance changes, and pseudo-label selection is often stabilized mainly by confidence without additional plausibility or structural constraints. PhysSANet addresses these limitations through four coupled components: PAPs for acquisition-aligned training views without requiring metadata or inversion, MHSAA for boundary/region modeling, an SSA-FPN for boundary-preserving multiscale fusion, and an MVCR scheme integrating teacher–student consistency with plausibility-aware filtering and structure-aware smoothness.

III. Methodology

A. Overview

Remote sensing CD is challenged by limited pixel-level annotations, acquisition-induced appearance variability, and the need for boundary-accurate delineation of man-made structures. Existing semi-supervised approaches are often limited by: 1) *acquisition-mismatched perturbations* that poorly reflect typical remote sensing appearance variations; 2) *structure-insensitive representation learning* that underutilizes boundaries and region coherence; and 3) *instability in consistency-based training*, where pseudo-labels can become unreliable under strong radiometric discrepancies. To address these limitations, we propose PhysSANet, a physics-inspired and structure-aware semi-supervised framework (Fig. 1); Algorithm 1 summarizes the training procedure.

PhysSANet comprises four tightly coupled components.

1) *PAP (Section III-B)* generates weak and strong views by approximating common acquisition-related effects, including atmospheric veiling (haze-like), illumination/exposure variation, and sensor noise. PAP perturbs a set of acquisition-related parameters within empirically reasonable ranges and further constrains them using simple image statistics (e.g., intensity percentiles and local contrast), thereby reducing out-of-distribution transformations that may corrupt consistency learning. This design provides acquisition-aligned training views without requiring per-image metadata or performing physical inversion.
2) *MHSAA (Section III-C)* enhances bitemporal features via three complementary branches: an edge-aware branch emphasizing geometric boundaries, a region-aware branch modeling spatial coherence of changes, and a cross-temporal branch capturing long-range correspondences between time steps. By explicitly encoding geometric cues, MHSAA attenuates responses to radiometric fluctuations while highlighting structurally meaningful discrepancies.
3) *SSA-FPN (Section III-D)* preserves fine-grained geometry during multiscale fusion. A spatial–structural attention block is embedded into a bidirectional feature pyramid: the top–down pathway propagates high-level semantics, while the bottom–up pathway refines localization; feature fusion at each scale is controlled by learnable weights. This design maintains sharp bound-

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Algorithm 1 Training Procedure of PhysSANet

Require: Labeled set $\mathcal{D}_L$, unlabeled set $\mathcal{D}_U$; student network f_{θ_s}, teacher network f_{θ_t}
Ensure: Trained student model f_{θ_s}

Initialize θ_s; set $\theta_t \leftarrow \theta_s$
for $t = 1$ to $T_{\max}$ **do**
  Sample a mini-batch $(x_L, y_L) \sim \mathcal{D}_L$, $x_U \sim \mathcal{D}_U$
  Compute supervised loss on labeled data: $\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{CE}}(f_{\theta_s}(x_L), y_L) + \mathcal{L}_{\text{Dice}}(f_{\theta_s}(x_L), y_L)$
  Generate physics-aware weak and strong perturbations of x_U using PAP: $\tilde{x}_w = \mathcal{P}(x_U; \alpha_{\text{weak}})$, $\tilde{x}_s = \mathcal{P}(x_U; \alpha_{\text{strong}})$
  Obtain teacher prediction with MHSAA and SSA-FPN: $\hat{y}_w = f_{\theta_t}^{\text{MHSAA+SSA-FPN}}(\tilde{x}_w)$
  Compute confidence and appearance-plausibility scores to build mask $\mathcal{M}$ (Section III-E)
  Obtain student prediction on strong perturbation: $\hat{y}_s = f_{\theta_s}^{\text{MHSAA+SSA-FPN}}(\tilde{x}_s)$
  Compute unsupervised loss on masked pixels: $\mathcal{L}_{\text{unsup}} = \mathcal{L}_{\text{pixel}} + \lambda_1 \mathcal{L}_{\text{spatial}} + \lambda_2 \mathcal{L}_{\text{cross}}$
  Update student parameters: $\theta_s \leftarrow \theta_s - \eta \nabla_{\theta_s}(\mathcal{L}_{\text{sup}} + \lambda_u(t)\mathcal{L}_{\text{unsup}})$
  Update teacher parameters by EMA: $\theta_t \leftarrow \mu\theta_t + (1-\mu)\theta_s$
end for
return f_{θ_s}

aries and thin objects across scales, which is critical for delineating building footprints and other man-made structures.

4) *MVCR (Section III-E)* extends the Mean Teacher paradigm with acquisition-aligned perturbations and structure-enhanced representations. Specifically, MVCR forms pseudo-labels from the teacher prediction on the weak view and applies a joint *confidence–plausibility* criterion to construct a reliable mask over unlabeled pixels. The plausibility score is computed in the observable image domain to quantify whether the strong view introduces abnormal local changes relative to the weak view (e.g., excessive intensity shifts or contrast disruption), thereby filtering pixels likely affected by severe appearance distortions. The student is then trained on the strong view with masked unsupervised losses, including pixelwise consistency, structure-aware spatial regularization, and cross-view/branch consistency, which jointly improve training stability under limited supervision.

Synergistic Design: PAP exposes the model to acquisition-related variability, encouraging MHSAA and SSA-FPN to learn structure-preserving representations that are less sensitive to radiometric changes. Conversely, the structure-aware modules promote geometrically coherent predictions, which help MVCR select more reliable pseudo-labels and reduce boundary artifacts. Together, this coupling between acquisition-aligned perturbations, structure-aware representation learning, and masked multiview consistency enables more effective utilization of unlabeled data than generic SSL baselines. A more explicit description of the interaction and information flow among the four components is provided in Section III-F.

B. Physics-Aware Perturbation

Most existing semi-supervised CD methods adopt generic natural-image augmentations, which can be mismatched with remote sensing acquisition variability and may degrade pseudo-label reliability. In practice, bitemporal remote sensing images often exhibit appearance variations induced by atmospheric veiling, illumination/exposure differences, and sensor noise. For commonly used high-resolution CD benchmarks, however, per-image metadata (e.g., acquisition geometry and atmospheric parameters) is typically unavailable, making fully parameterized physics-based simulation impractical and difficult to reproduce.

To address this issue, we propose a *physics-inspired, acquisition-aligned perturbation* strategy termed *PAP*. Here, "physics-aware" indicates that PAP is *formation-inspired* and *parameterized* by common acquisition-related factors (veiling, illumination/exposure, and noise), rather than being a generic photometric jitter. Importantly, PAP is *metadata-free* and *noninversion*: it does not estimate physical variables (e.g., atmospheric parameters) and does not aim at physically exact simulation. Instead, it samples perturbation parameters from empirically chosen ranges and enforces explicit *observable-domain plausibility constraints* (e.g., percentile/contrast bounds and a similarity safeguard), so that the perturbed image stays within a reproducible and verifiable appearance envelope while preserving semantic content and change patterns. As shwon in Fig. 2, PAP generates weak and strong perturbation views by combining haze-like veiling, illumination/exposure variation, and sensor noise while preserving the underlying change patterns. Given an input image I, PAP is defined as a lightweight composition of three operators

$$\tilde{I} = \mathcal{P}(I;\alpha) = \mathcal{N}(\mathcal{L}(\mathcal{H}(I;\theta_h);\theta_l);\theta_n) \tag{1}$$

where $\mathcal{H}(\cdot;\theta_h)$ simulates haze-like veiling, $\mathcal{L}(\cdot;\theta_l)$ simulates illumination/exposure differences, and $\mathcal{N}(\cdot;\theta_n)$ simulates sensor noise. All parameters are sampled from empirically chosen ranges and then adjusted by statistic-based constraints to reduce implausible radiometric drift.

1) Haze-Like Veiling: We simulate haze-like appearance degradation using a formation-inspired convex mixing

$$\mathcal{H}(I;\theta_h) = tI + (1-t)A \tag{2}$$

where $\theta_h = \{t, A\}$, $t \in [0,1]$ controls the veiling strength, and A denotes an airlight term. Since absolute atmospheric conditions are unknown, t is sampled from an empirically chosen range and then constrained using image statistics, e.g., by bounding global contrast changes or limiting intensity percentiles of $\mathcal{H}(I;\theta_h)$ relative to the input. The airlight term A is instantiated as a simple proxy derived from high-intensity statistics of the input image (e.g., channelwise top percentiles), thereby avoiding additional learnable parameters or metadata dependencies while encouraging plausible brightening without abnormal color shifts. In all experiments, the veiling coefficient t is sampled from $[0.75, 0.90]$ for weak views and $[0.40, 0.70]$ for strong views. The airlight term A is estimated from the channelwise 99th intensity percentile of the input image, which provides a simple and reproducible proxy for bright-scattering effects without introducing metadata dependence.

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2) Illumination and Exposure Differences: To mimic illumination/exposure differences between temporal observations, we apply a parametric intensity transformation

$$\mathcal{L}(I;\theta_l) = \text{clip}(gI^{\gamma} + b) \tag{3}$$

where $\theta_l = \{g, \gamma, b\}$, and g, γ, and b control global gain, nonlinearity, and bias, respectively. Compared with generic brightness or gamma jitter, the parameters (g, γ, b) are *jointly* sampled from empirically chosen ranges and then constrained by histogram statistics to reduce excessive saturation, contrast collapse, or unrealistic tonal shifts that may distort semantic content. The clip$(\cdot)$ operator ensures a valid intensity range. Specifically, the gain g, gamma γ, and bias b are sampled from $g \in [0.8, 1.2]$, $\gamma \in [0.7, 1.4]$, and $b \in [-0.05, 0.05]$, respectively. These parameters are jointly constrained by histogram statistics to reduce excessive saturation, contrast collapse, and unrealistic tonal shifts.

3) Sensor Noise: We model noise as the combination of signal-independent and signal-dependent-like components

$$\mathcal{N}(I;\theta_n) = I + \epsilon_G + \epsilon_P \tag{4}$$

where θ_n parameterizes noise levels, $\epsilon_G \sim \mathcal{N}(0, \sigma^2)$ denotes the Gaussian noise, and ϵ_P acts as a simple proxy for heteroscedastic noise via intensity-dependent scaling with normalized intensity, while remaining conservatively bounded to preserve structural and change-related information. Noise levels are determined using normalized intensity statistics and clipped to reduce the risk of corrupting fine structures. In all experiments, the Gaussian noise standard deviation is sampled from $\sigma \in [0.01, 0.04]$. The intensity-dependent component is defined as $\epsilon_P = \rho \cdot I_{\text{norm}}$, where $I_{\text{norm}} \in [0, 1]$ denotes the normalized image intensity and $\rho \in [0.0, 0.02]$ is the scaling factor. To avoid corrupting thin structures and small change regions, the noise terms are conservatively bounded, with each component clipped to $[-0.05, 0.05]$.

4) Weak and Strong Views With Strength Control: PAP generates weak and strong views via a scalar strength parameter $\alpha \in [0, 1]$. Let $I_{\text{pap}} = \mathcal{N}(\mathcal{L}(\mathcal{H}(I;\theta_h);\theta_l);\theta_n)$ denote the perturbation result (before strength control). We define

$$\mathcal{P}(I;\alpha) = (1-\alpha)I + \alpha I_{\text{pap}} \tag{5}$$

so that larger α yields stronger appearance variations. Weak perturbations stabilize teacher predictions, while strong perturbations provide challenging inputs for student learning. To reduce semantic drift, we apply a conservative plausibility safeguard in the observable image domain. Specifically, after obtaining $\tilde{I} = \mathcal{P}(I;\alpha)$, we enforce simple statistic-based constraints and resample perturbation parameters (or shrink α) if they are violated

$$\begin{aligned} \text{accept}(\tilde{I}) \Leftrightarrow & \max_{p\in\Omega} \left| Q_p(\tilde{I}) - Q_p(I) \right| \le \epsilon_p \\ & \left| C(\tilde{I}) - C(I) \right| \le \epsilon_c \\ & \text{Sim}(I, \tilde{I}) \ge \epsilon_s \end{aligned} \tag{6}$$

where $Q_p(\cdot)$ denotes the intensity percentiles with $\Omega = 1, 99$, $C(\cdot)$ is the standard deviation of intensity, and Sim$(\cdot)$ is structural similarity index (SSIM).

In summary, PAP provides acquisition-aligned perturbations for semi-supervised consistency learning while remaining applicable to datasets without physical metadata. Unlike generic photometric augmentations (e.g., color jitter or RandAugment), PAP defines a parameterized family of acquisition-aligned perturbations subject to observable-domain plausibility constraints. The parameter ranges are chosen to reflect common radiometric variations in high-resolution optical remote sensing, including haze-like veiling, illumination/exposure changes, and sensor noise, and are further adapted using simple per-image statistics. To avoid excessive saturation, contrast distortion, and structural drift, we impose a percentile clamp, a contrast bound, and a similarity safeguard. In all experiments, the thresholds are fixed to $\epsilon_p = 0.15$, $\epsilon_c = 0.10$, and $\epsilon_s = 0.85$. These values are determined using a conservative two-stage procedure: feasible ranges are first specified from simple image statistics to keep perturbed views within a plausible appearance envelope, and the final thresholds are then selected on the validation set to balance perturbation diversity and structural fidelity. The same configuration is used across all three benchmarks without dataset-specific retuning. When the safeguard is violated, perturbation parameters are resampled up to $K = 3$ times; if the constraints are still unmet, α is reduced by 0.1 until acceptance.

C. Multihead Structure-Aware Attention

High-resolution remote sensing imagery exhibits strong geometric regularities (e.g., building edges and road boundaries) but also large photometric variations caused by illumination, shadows, or seasonal tone shifts. Conventional attention mechanisms treat all feature discrepancies equally and, therefore, may respond strongly to radiometric noise while underemphasizing stable structural cues. To address this issue, we introduce a lightweight *MHSAA* module [Fig. 1(b)] that explicitly incorporates geometric priors into bitemporal feature interaction.

MHSAA consists of three complementary branches. *Edge-aware attention* extracts gradient maps

$$\mathbf{E}^{t_i} = \sqrt{\left(\text{Sobel}_x\left(\mathbf{F}^{t_i}\right)\right)^2 + \left(\text{Sobel}_y\left(\mathbf{F}^{t_i}\right)\right)^2} \tag{7}$$

and computes attention on these boundary-enhanced features

$$\text{EdgeAttn}^h = \text{softmax}\left(\frac{\mathbf{Q}_e^h \left(\mathbf{K}_e^h\right)^T}{\sqrt{d_k}}\right)\mathbf{V}_e^h \tag{8}$$

where $\mathbf{Q}_e^h = \mathbf{E}^{t_1}\mathbf{W}_{Q_e}^h$ and $\mathbf{K}_e^h = \mathbf{E}^{t_2}\mathbf{W}_{K_e}^h$. This allows the network to focus on structural discontinuities such as building outlines while being insensitive to illumination changes.

Structure-aware attention operates on absolute difference features $\mathbf{D} = |\mathbf{F}^{t_1} - \mathbf{F}^{t_2}|$

$$\text{StructAttn}^h = \text{softmax}\left(\frac{\mathbf{Q}_d^h \left(\mathbf{K}_d^h\right)^T}{\sqrt{d_k}}\right)\mathbf{V}_d^h \tag{9}$$

to capture region-level homogeneous changes, improving coherence within newly constructed or demolished areas.

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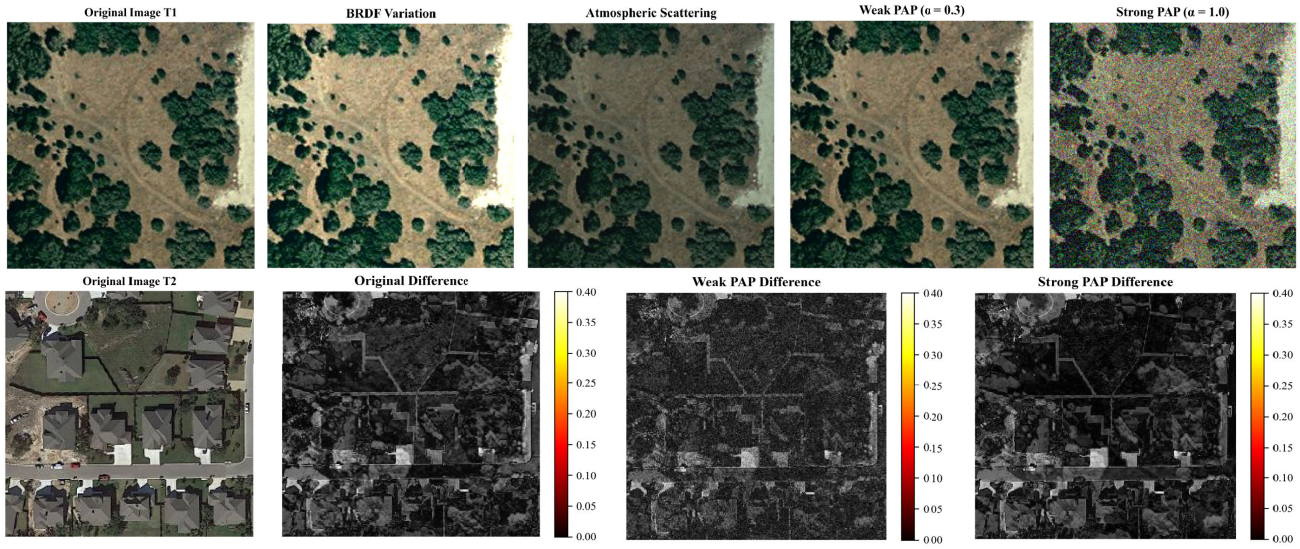


Fig. 2. PAP module for remote sensing CD. (First row) Individual physics effects demonstrate the modular design, where haze-like veiling simulates visibility degradation and visibility variations, and illumination-related variation approximates appearance changes induced by differences in acquisition conditions. Combined perturbations (weak and strong PAP) show controllable augmentation strength through parameter α. (Second row) CD difference maps reveal that PAP preserves semantic change information while introducing realistic variations. Strong perturbations provide more diverse augmentations for robust training while maintaining the underlying change patterns.

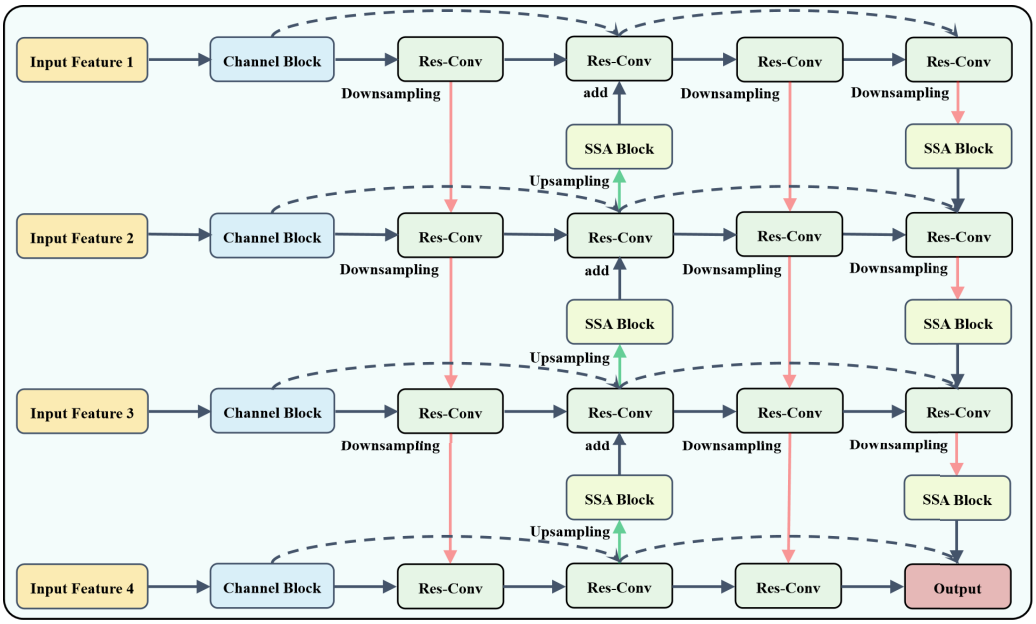


Fig. 3. Architecture of the SSA-FPN. The framework integrates coordinate attention mechanisms with bidirectional feature pyramid structures to preserve spatial information while enabling effective multiscale feature fusion for accurate CD.

Cross-temporal attention directly matches raw features

$$\text{CrossAttn}^h = \text{softmax}\left(\frac{\mathbf{Q}^{t_1,h}\left(\mathbf{K}^{t_2,h}\right)^T}{\sqrt{d_k}}\right)\mathbf{V}^{t_2,h} \tag{10}$$

to capture semantic correspondences, helping suppress false positives (FPs) arising from seasonal or appearance variations. The three outputs are fused through pixelwise adaptive weights

$$\boldsymbol{\alpha} = \text{softmax}\left(\text{MLP}\left(\left[\mathbf{A}_{\text{edge}}, \mathbf{A}_{\text{struct}}, \mathbf{A}_{\text{cross}}\right]\right)\right) \tag{11}$$

yielding enhanced features

$$\mathbf{F}_{\text{enhanced}} = \sum_{i=1}^{3} \alpha_i \mathbf{A}_i + \mathbf{F}^{t_1}. \tag{12}$$

This design enables the network to emphasize edges in boundary-rich regions, different features in homogeneous change areas, and temporal matching in illumination-varying regions. The ablation results further show that MHSAA provides a +1.0% intersection over union (IoU) gain on LEVIR-CD and up to +2.6% on WHU-CD, confirming that explicit geometric modeling complements PAP-induced radiometric robustness.

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D. Spatial–Structural Attention-Enhanced Feature Pyramid Network

FPNs are widely used for multiscale fusion, yet standard variants (FPN, PANet, and BiFPN) tend to blur spatial details and lack explicit structural modeling—both crucial for delineating fine-grained change boundaries in high-resolution remote sensing. We propose the *SSA-FPN* (Fig. 3), which augments a bidirectional pyramid with a lightweight spatial–structural attention block to preserve geometric fidelity during multiscale fusion and to provide structure-aware features for both MHSAA and MVCR.

1) Spatial–Structural Attention Block: Given an input feature $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, SSA jointly encodes directional spatial context and boundary cues. Horizontal and vertical contexts are obtained via coordinate pooling

$$\mathbf{G}_{h,w} = \mathrm{Conv}_{1\times 1}\left(\mathrm{AvgPool}_{w,h}(\mathbf{X})\right) \tag{13}$$

where $\mathrm{AvgPool}_w$ and $\mathrm{AvgPool}_h$ aggregate features along width and height, capturing long-range dependencies along road-like and building-row structures. In parallel, we extract edge-aware structural responses

$$\mathbf{E} = \sqrt{(\mathrm{Sobel}_x\mathbf{X})^2 + \left(\mathrm{Sobel}_y\mathbf{X}\right)^2}, \quad \mathbf{G}_e = \mathrm{Conv}_{3\times 3}(\mathrm{BN}(\mathbf{E})) \tag{14}$$

so that the contours of buildings and man-made objects are highlighted. The three cues are fused to refine the feature map

$$\mathrm{SSA}(\mathbf{X}) = \mathbf{X} \odot \sigma\left(\mathrm{Conv}_{1\times 1}([\mathbf{G}_h \oplus \mathbf{G}_w \oplus \mathbf{G}_e])\right) \tag{15}$$

where $\oplus$ denotes the concatenation, σ is the sigmoid function, and $\odot$ is the elementwise multiplication. This behaves as a boundary-aware coordinate attention module that enhances change-relevant responses while suppressing background clutter.

2) Bidirectional Multiscale Fusion: SSA blocks are integrated into a bidirectional pyramid similar in spirit to BiFPN [39], but adapted to the bitemporal CD setting. Encoder features at four scales (1/4–1/32 resolution) are first projected to a unified channel dimension (256) by 1×1 gating layers. The top–down path propagates high-level semantics

$$\mathbf{F}_i^{td} = \mathrm{Upsample}\left(\mathrm{SSA}\left(\mathbf{F}_{i+1}^{td}\right)\right) \tag{16}$$

while the bottom–up path restores localization details

$$\mathbf{F}_i^{bu} = \mathrm{Downsample}\left(\mathbf{F}_{i-1}^{out}\right). \tag{17}$$

At each level i, three sources (input, top–down, and bottom–up) are fused with learnable nonnegative weights

$$\mathbf{F}_i^{fuse} = \frac{w_1\mathrm{Gate}\left(\mathbf{F}_i^{in}\right) + w_2\mathbf{F}_i^{td} + w_3\mathbf{F}_i^{bu}}{w_1 + w_2 + w_3 + \epsilon} \tag{18}$$

followed by an SSA refinement:

$$\mathbf{F}_i^{out} = \mathrm{SSA}\left(\mathbf{F}_i^{fuse}\right). \tag{19}$$

Here, $w_1, w_2, w_3 \geq 0$ are the trainable scalars initialized to 1.0 and $\epsilon = 10^{-4}$ ensures numerical stability.

This design enforces structural consistency across pyramid levels—edge responses ($\mathbf{G}_e$) preserve sharp boundaries, while coordinate context ($\mathbf{G}_h, \mathbf{G}_w$) maintains long-range spatial relationships. The ablation results further show that SSA-FPN improves IoU by +0.9% on LEVIR-CD and +4.3% on WHU-CD, with larger gains on complex, irregular building layouts. Meanwhile, the added computational cost is modest compared with the backbone, making SSA-FPN a practical drop-in replacement for standard FPNs in CD.

E. Multiview Consistency Regularization

In semi-supervised CD, confidence-only pseudo-label filtering is often insufficient under strong acquisition-related appearance variations (e.g., atmospheric veiling, illumination/exposure differences, and sensor noise). Moreover, structural ambiguity at thin boundaries and small objects can further amplify pseudo-label noise. Without relying on per-image metadata, we design MVCR as an *acquisition-plausible/observable-domain-plausible multiview regularization* mechanism that integrates teacher–student consistency, appearance-plausibility-aware pseudo-label filtering in the observable image domain, and structure-aware smoothness for boundary preservation. Fig. 4 illustrates the difference between MVCR and representative consistency regularization strategies.

1) Teacher–Student Consistency With Weak and Strong Views: MVCR follows the Mean Teacher paradigm [27]. For each unlabeled bitemporal pair $x = (x^{t1}, x^{t2})$, we generate two views via PAP

$$\tilde{x}_w = \mathcal{P}(x; \alpha_{\mathrm{weak}}), \quad \tilde{x}_s = \mathcal{P}\left(x; \alpha_{\mathrm{strong}}\right) \tag{20}$$

where α_{weak} produces a stable teacher input, and α_{strong} produces a more challenging student input. The teacher makes predictions on the weak view

$$p_t = \sigma\left(f_{\theta_t}(\tilde{x}_w)\right) \tag{21}$$

and the student predicts on the strong view

$$p_s = \sigma\left(f_{\theta_s}(\tilde{x}_s)\right). \tag{22}$$

The teacher is updated by an EMA

$$\theta_t \leftarrow \mu\,\theta_t + (1-\mu)\,\theta_s \tag{23}$$

with momentum $\mu = 0.999$.

2) Appearance-Plausibility-Aware Pseudo-Label Selection: To suppress pseudo-labels corrupted by overly strong perturbations, we compute an appearance plausibility map in the observable image domain to down-weight pixels where the strong view introduces abnormal radiometric distortions relative to the weak view. The key idea is that a valid strong view should change appearance while largely preserving local structural content. To operationalize this requirement without metadata or physical inversion, we adopt an *observable-domain plausibility proxy* that jointly measures: 1) radiometric drift and 2) structural fidelity. Specifically, luminance difference captures global/low-frequency intensity shifts that commonly arise from exposure, gamma, and bias changes, while luminance-gradient difference captures local contrast/edge-strength deviations, serving as a proxy for whether the perturbation excessively disrupts structural cues. This design is consistent with the qualitative effects of typical

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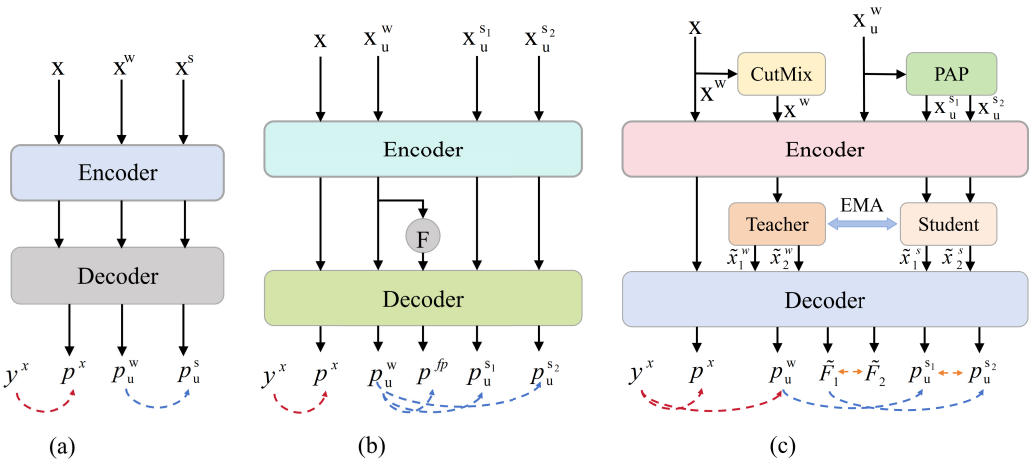


Fig. 4. Comparison of consistency regularization strategies in semi-supervised CD. (a) FixMatch applies a generic weak–strong augmentation without domain-specific considerations. (b) UniMatch employs dual-branch consistency but lacks physics-inspired constraints. (c) PhysSANet/MVCR framework integrates PAPs and structure-aware consistency to generate more reliable pseudo-labels for building CD.

acquisition variations: for example, haze-like veiling can be expressed by a convex mixing model [see (2)], whose dominant observable effects include brightness shifts and contrast compression (hence, changes in luminance and gradients), and illumination/exposure variation [see (3)] also primarily manifests as intensity and contrast changes. Therefore, the gradient-plus-luminance proxy provides a conservative plausibility check in the observable image domain, rather than a claim of physical correctness.

a) Luminance and gradient for RGB bitemporal pairs: For an RGB image $I \in \mathbb{R}^{H\times W\times 3}$, we compute luminance as

$$\ell(I) = \sum_c w_c I_c, \quad \sum_c w_c = 1 \tag{24}$$

where $\ell(\cdot)$ denotes the luminance extraction operator. We then apply the Sobel operator ∇ to $\ell(\cdot)$ and use $\|\nabla\ell(\cdot)\|_2$ to denote the corresponding per-pixel gradient magnitude. For a bitemporal pair $\tilde{x} = (\tilde{x}^{t_1}, \tilde{x}^{t_2})$, we define the gradient proxy as the average over two timestamps

$$\begin{aligned} g(\tilde{x}) &= \frac{1}{2}\left(\|\nabla\ell\left(\tilde{x}^{t_1}\right)\|_2 + \|\nabla\ell\left(\tilde{x}^{t_2}\right)\|_2\right) \\ \ell(\tilde{x}) &= \frac{1}{2}\left(\ell\left(\tilde{x}^{t_1}\right) + \ell\left(\tilde{x}^{t_2}\right)\right). \end{aligned} \tag{25}$$

b) Observable-domain plausibility map: We define a plausibility map $S_{\text{pl}} \in [0,1]^{H\times W}$

$$\begin{aligned} S_{\text{pl}}(i,j) = \exp\left(-\beta\,|g(\tilde{x}_w)(i,j) - g(\tilde{x}_s)(i,j)|\right) \\ \cdot \exp\left(-\delta\,|\ell(\tilde{x}_w)(i,j) - \ell(\tilde{x}_s)(i,j)|\right) \end{aligned} \tag{26}$$

where $\beta, \delta > 0$ control sensitivity and are set to $\beta = 10$ and $\delta = 5$ for all experiments after normalizing $\ell(\cdot)$ and the Sobel magnitude to $[0,1]$ per image, so that deviations of 0.1–0.15 produce a noticeable down-weighting. S_{pl} down-weights pixels where the strong view introduces excessive local-contrast changes or large luminance shifts relative to the weak view, which are more likely to indicate perturbation-induced semantic drift or degraded correspondence.

c) Joint confidence–plausibility masking: We construct the pseudo-label selection mask by jointly thresholding teacher confidence and plausibility

$$\mathcal{M} = \left\{(i,j) \mid \max_c p_t^{(c)}(i,j) \geq \tau_c \ \wedge \ S_{\text{pl}}(i,j) \geq \tau_p\right\} \tag{27}$$

and use the equivalent binary mask $M(i,j) = \mathbf{1}[(i,j) \in \mathcal{M}]$ for loss computation. We apply this safeguard only to the strong view to prevent overly aggressive perturbations, while the weak view uses mild ranges by design. The hard pseudo-label is $\hat{y}(i,j) = \arg\max_c p_t^{(c)}(i,j)$.

3) Hierarchical Consistency With Structure-Aware Smoothness: MVCR enforces complementary constraints: masked teacher–student agreement, cross-view consistency, and structure-aware smoothness.

a) Masked pixel-level consistency: We minimize a masked kullback-leibler (KL) divergence between teacher and student predictions

$$\mathcal{L}_{\text{pixel}} = \frac{1}{\sum_{i,j} M(i,j)} \sum_{i,j} M(i,j)\ \text{KL}\big(p_t(i,j)\ \|\ p_s(i,j)\big). \tag{28}$$

b) Cross-view consistency: We generate two independent strong views $\tilde{x}_{s_1}, \tilde{x}_{s_2}$ and enforce agreement between the corresponding student predictions using a symmetric temperature-scaled KL divergence

$$\begin{aligned} \mathcal{L}_{\text{cross}} = \frac{1}{HW}\sum_{i,j}\frac{1}{2}\Big[&\text{KL}\big(\sigma(z_{s_1}(i,j)/T)\ \|\ \sigma(z_{s_2}(i,j)/T)\big) \\ &+ \text{KL}\big(\sigma(z_{s_2}(i,j)/T)\ \|\ \sigma(z_{s_1}(i,j)/T)\big)\Big] \end{aligned} \tag{29}$$

where z_{s_1} and z_{s_2} are the student logits, and T is the temperature.

c) Structure-aware smoothness with boundary preservation: To suppress high-frequency noise while preserving boundaries, we introduce an edge-aware smoothness term

$$\mathcal{L}_{\text{spatial}} = \frac{1}{\sum_{i,j} M(i,j)} \sum_{i,j} M(i,j)\ \|\nabla p_s(i,j)\|_2^2 \cdot w_{\text{edge}}(i,j) \tag{30}$$

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with

$$w_{\text{edge}}(i,j) = \exp\left(-\kappa\, g\left(\tilde{x}_w\right)(i,j)\right) \tag{31}$$

where $\kappa > 0$ reduces over-smoothing near strong edges and is set to $\kappa = 5$ for all experiments; using $\tilde{x}_w$ provides a more stable edge cue than $\tilde{x}_s$ under strong perturbations.

4) Final Objective:

$$\mathcal{L}_{\text{unsup}} = \mathcal{L}_{\text{pixel}} + \lambda_1 \mathcal{L}_{\text{spatial}} + \lambda_2 \mathcal{L}_{\text{cross}} \tag{32}$$

and the total training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda_u(t)\ \mathcal{L}_{\text{unsup}}. \tag{33}$$

a) Remark: MVCR does not assume access to physical variables and does not attempt physical inversion; the plausibility term is purely an *observable-domain* appearance check rather than a physical correctness test. It improves pseudo-label reliability by filtering unreliable regions using appearance plausibility in the observable image domain and by enforcing structure-aware smoothness for boundary preservation, jointly stabilizing consistency learning under remote-sensing-specific appearance variations.

F. Component Interaction and Information Flow

The four components of PhysSANet form a connected pipeline for semi-supervised bitemporal learning. Given an unlabeled input pair $x = (x^{t_1}, x^{t_2})$, PAP generates a weak view $\tilde{x}_w$ and a strong view $\tilde{x}_s$ representing two perturbed observations of the same scene. Both views are processed by the shared Siamese encoder to extract bitemporal features, which are then refined by MHSAA to capture boundary cues, regional structure, and cross-temporal correspondences, and fused by SSA-FPN across multiple scales into structure-preserving representations. Based on these representations, the teacher predicts on $\tilde{x}_w$, and the student predicts on $\tilde{x}_s$, while MVCR constructs masked consistency supervision from the resulting teacher–student pair.

PAP plays a coupling role within this pipeline. By perturbing the observation conditions of the bitemporal pair without altering the underlying semantic change pattern, PAP enables MHSAA and SSA-FPN to operate on inputs that reflect realistic acquisition-related differences, including illumination shifts, haze-like veiling, and sensor noise. This encourages the representation learning modules to capture stable structural cues rather than acquisition-specific appearance details, thereby coupling the perturbation modeling stage with the feature extraction stage.

The representations produced by MHSAA and SSA-FPN provide the basis for consistent supervision in MVCR. MHSAA contributes boundary- and correspondence-aware feature refinement, and SSA-FPN contributes multiscale spatial aggregation that preserves thin and small structures. Together, they provide more spatially coherent predictions on the weak view, on which MVCR builds its confidence- and plausibility-guided pseudo-label masks. The overall interaction, thus, follows a complete flow: from acquisition-aligned perturbation, through structure-aware representation learning, to masked consistency optimization.

TABLE II
DATA PARTITIONS UNDER DIFFERENT ANNOTATION RATIOS IN THE SEMI-SUPERVISED SETTING

Dataset	Ratio	Labeled	Unlabeled	Val	Test
LEVIR-CD	5%	356	6,764	1,024	2,048
	10%	712	6,408	1,024	2,048
	20%	1,424	5,696	1,024	2,048
	40%	2,848	4,272	1,024	2,048
WHU-CD	5%	297	5,650	743	744
	10%	594	5,353	743	744
	20%	1,189	4,758	743	744
	40%	2,378	3,569	743	744
SYSU-CD	5%	600	11,400	4,000	4,000
	10%	1,200	10,800	4,000	4,000
	20%	2,400	9,600	4,000	4,000
	40%	4,800	7,200	4,000	4,000

IV. EXPERIMENTS

We evaluate PhysSANet on three benchmark datasets under varying annotation ratios, compare it with state-of-the-art semi-supervised and fully supervised methods, and conduct ablation studies to assess the contribution of each component.

A. Datasets

We consider three widely used high-resolution CD datasets: LEVIR-CD, WHU-CD, and SYSU-CD. All images are tiled into nonoverlapping 256 × 256 patches for training and evaluation.

LEVIR-CD [40] is an urban building CD dataset built from very high-resolution Google Earth imagery (0.5 m/pixel). After tiling, we obtain 10 240 bitemporal pairs, split into 7120/1024/2048 for training/validation/testing, respectively. The dataset focuses on building changes across 20 urban regions in Texas, characterized by clear boundaries and consistent structural patterns.

WHU-CD [41] contains large-scale aerial image pairs (0.2 m/pixel) depicting building changes in Christchurch, New Zealand. We generate 7487 patches, with 5947 for training, 743 for validation, and 744 for testing. The data capture multiscale building changes before and after the 2011 earthquake under varying imaging conditions.

SYSU-CD [42] consists of 20 000 bitemporal image pairs (0.5 m/pixel) over Hong Kong, covering diverse change types such as building construction/demolition, road development, vegetation dynamics, and water variations. We follow the common 12 000/4000/4000 split. Its heterogeneous land-cover composition makes SYSU-CD suitable for evaluating cross-scene generalization.

In the semi-supervised setting, we follow the protocol in [29] and randomly sample a subset of the training set as labeled, with labeling ratios of 5%, 10%, 20%, and 40%. The remaining training samples are treated as unlabeled and used for unsupervised consistency regularization and pseudo-label learning. Validation and test sets are kept fixed across all annotation ratios. The detailed data partitions under each ratio are summarized in Table II.

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B. Implementation Details

All models are implemented in PyTorch and trained on a workstation equipped with two NVIDIA RTX 4090 GPUs (24-GB memory each). Input bitemporal images are resized and cropped to 256 × 256 pixels. Unless otherwise stated, we apply standard spatial augmentations (random horizontal/vertical flips and 90° rotations) in conjunction with the proposed PAPs.

PAP Configuration: PAP is configured as a reproducible, metadata-free perturbation strategy. All parameters are sampled from typical ranges and then constrained using image statistics to reduce semantic drift. Specifically, the haze-like veiling in (2) samples t from weak/strong ranges for teacher/student views, and sets A using high-intensity percentiles of the input image. The illumination and exposure transform in (3) samples (g, γ, b) and applies histogram-based constraints to prevent over-saturation and excessive darkening. The noise model in (4) uses intensity-normalized noise levels with conservative bounds to preserve structural content. The perturbation strength α is sampled from $[0.1, 0.3]$ for weak views and $[0.5, 0.8]$ for strong views, and we use a short warm-up stage with a smaller α to stabilize early pseudo-labels. Importantly, PAP does not require per-image metadata and does not attempt to recover physical quantities; its goal is to mimic common acquisition-related appearance variations for robust semi-supervised training. Unless otherwise stated, all PAP parameter ranges and plausibility-safeguard thresholds described in Section III-B are kept strictly fixed across all three benchmark datasets without any dataset-specific tuning, ensuring reproducibility and fair comparison.

Network Configuration: Both student and teacher networks adopt the same backbone, augmented with the proposed MHSAA and SSA-FPN modules. MHSAA uses eight attention heads with key dimension $d_k = 64$ for each branch (edge-aware, structure-aware, and cross-temporal), leading to a 512-D fused representation after concatenation. SSA-FPN operates on four pyramid levels corresponding to encoder stages at 1/4, 1/8, 1/16, and 1/32 of the input resolution, with all feature maps projected to 256 channels via 1 × 1 gating convolutions. Dropout with a rate of 0.1 is applied to attention modules during training and disabled during inference.

Training Strategy and Optimization: We follow the Mean Teacher paradigm [27]. The teacher parameters are updated as an EMA of the student parameters with momentum $\mu = 0.999$, providing temporally smoothed targets. In each iteration, the teacher processes weakly perturbed inputs to produce pseudo-labels, and the student is trained on strongly perturbed inputs using both supervised and unsupervised losses.

The total loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda_u(t)\left[\mathcal{L}_{\text{pixel}} + 0.1\,\mathcal{L}_{\text{spatial}} + 0.1\,\mathcal{L}_{\text{cross}}\right] \tag{34}$$

where $\mathcal{L}_{\text{sup}}$ is the supervised loss on labeled data, defined as

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{CE}}\left(f_{\theta_s}(x_L), y_L\right) + \mathcal{L}_{\text{Dice}}\left(f_{\theta_s}(x_L), y_L\right). \tag{35}$$

The unsupervised weight is gradually increased according to

$$\lambda_u(t) = \lambda_{\max} \cdot \text{sigmoid}\left(\frac{t - t_0}{\tau}\right) \tag{36}$$

with $\lambda_{\max} = 1.0$, $t_0 = 1000$, and $\tau = 200$, so that the contribution of unlabeled data is smoothly ramped up during the first training stages.

For MVCR, pseudo-labels are filtered using a confidence threshold $\tau_c = 0.95$ and an *appearance-plausibility* threshold $\tau_p = 0.8$ (Section III-E). For the observable-domain plausibility map in (26), we normalize $\ell(\cdot)$ and the Sobel magnitude to $[0, 1]$ per image and set $\beta = 10$ and $\delta = 5$ for all datasets. With this normalization, β and δ are chosen so that moderate deviations (e.g., 0.1–0.15 in normalized gradient/luminance) lead to a noticeable exponential down-weighting, thereby conservatively filtering pixels likely affected by excessive perturbation-induced radiometric drift or contrast disruption. For cross-view consistency in (29), we use temperature $T = 1.0$. For the edge-aware weight in (31), we set $\kappa = 5$, and compute $g(\tilde{x}_w)$ from the weak view for stability under strong perturbations.

We use AdamW for optimization with a learning rate of 3×10^{-4}, a weight decay 5×10^{-4}, and a polynomial learning rate decay schedule with power 0.9. Unless stated otherwise, models are trained for 40 000 iterations with a batch size of 16 per GPU (eight labeled and eight unlabeled pairs), i.e., 32 in total across two GPUs. All experiments are repeated five times with different random seeds, and we report the mean performance.

C. Evaluation Metrics

We evaluate all methods using five commonly adopted metrics for binary CD: precision (P), recall (R), $F1$-score ($F1$), IoU, and panoptic quality (PQ) [43]. Precision, recall, and $F1$-score are defined as

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R} \tag{37}$$

where TP, FP, and FN denote the number of true positive, false positive, and false negative change pixels, respectively. Precision reflects the reliability of detected changes, recall measures detection completeness, and $F1$ provides a balanced summary. The IoU (Jaccard index) is given by

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{38}$$

and quantifies the overlap between predicted and ground-truth change regions, serving as a stringent measure of segmentation quality (SQ). To further assess structural fidelity, we compute PQ, which decomposes into SQ and recognition quality (RQ) [43]

$$PQ = \underbrace{\frac{\sum_{(p,g)\in TP} \text{IoU}(p, g)}{|TP|}}_{\text{SQ}} \times \underbrace{\frac{|TP|}{|TP| + \frac{1}{2}|FP| + \frac{1}{2}|FN|}}_{\text{RQ}} \tag{39}$$

where (p, g) denotes the matched prediction–ground-truth instance pairs. In our setting, instances are defined as connected components in the binary change maps. PQ is, therefore, particularly informative for evaluating the geometry and completeness of individual change objects, which directly relates to the structure-aware design of PhysSANet.

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TABLE III

COMPREHENSIVE PERFORMANCE COMPARISON ON LEVIR-CD DATASET UNDER VARYING ANNOTATION RATIOS (%). BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED. ALL VALUES REPRESENT MEAN OVER FIVE RUNS WITH DIFFERENT RANDOM SEEDS

Method	5%					10%					20%					40%				
	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ
S4GAN	64.0	88.2	72.8	79.8	58.1	67.0	89.8	74.5	81.4	60.5	73.4	92.1	79.2	85.1	64.2	76.6	94.2	80.3	86.7	66.4
SemiCDNet	67.6	87.5	76.2	81.5	61.2	71.5	89.3	79.8	84.3	63.8	74.3	90.9	81.5	85.9	65.4	76.7	91.8	82.3	86.8	66.6
SemiCD	72.5	88.1	82.2	85.0	65.8	75.5	89.6	84.5	87.0	68.2	76.2	90.4	85.8	88.0	69.1	79.4	91.8	85.5	88.5	70.3
FixMatch	75.8	88.5	84.1	86.2	67.2	77.9	90.2	85.8	87.9	69.8	79.6	91.6	86.2	88.8	71.5	81.2	92.4	87.0	89.6	72.8
UniMatch	80.7	91.2	88.5	89.8	71.8	82.0	92.4	89.1	90.7	73.5	82.7	92.8	89.3	91.0	74.8	83.0	93.5	88.0	90.7	73.9
C2F-SemiCD	81.2	93.1	86.5	89.7	72.3	82.1	**93.8**	87.2	90.4	73.9	82.8	93.5	88.1	90.7	74.5	83.7	**94.1**	88.5	91.2	76.2
ISCDNet	81.8	**94.7**	85.8	90.0	72.5	82.3	93.1	88.6	90.8	73.7	82.5	**93.9**	87.2	90.4	73.6	83.6	93.2	89.1	91.1	75.9
PhysSANet	**81.9**	91.8	**89.2**	**90.5**	**72.8**	**82.5**	92.8	**89.8**	**91.2**	**74.2**	**83.0**	93.2	**90.1**	**91.6**	**75.0**	**83.9**	93.5	**89.5**	**91.8**	**76.6**
SupBaseline	69.3	86.9	80.5	83.6	62.1	76.0	90.1	84.2	87.0	67.8	77.6	91.5	86.6	89.0	70.5	81.2	92.6	87.8	90.1	72.8

TABLE IV

COMPREHENSIVE PERFORMANCE COMPARISON ON WHU-CD DATASET UNDER VARYING ANNOTATION RATIOS (%). BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED. ALL VALUES REPRESENT MEAN OVER FIVE RUNS WITH DIFFERENT RANDOM SEEDS

Method	5%					10%					20%					40%				
	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ
S4GAN	58.3	78.2	72.5	75.2	58.1	62.6	81.1	75.8	78.3	62.5	70.8	85.8	81.1	83.4	66.8	79.0	88.8	87.9	88.3	69.8
SemiCDNet	61.7	76.5	78.2	77.3	61.2	65.0	78.8	81.5	80.1	63.8	71.7	82.1	85.2	83.6	65.4	77.1	85.3	88.3	86.8	67.0
SemiCD	65.8	82.1	79.8	80.9	65.8	68.1	84.5	80.9	82.7	68.2	74.8	87.2	84.5	85.8	69.1	80.4	90.5	87.8	89.1	71.7
FixMatch	76.8	85.2	88.5	86.8	69.5	79.5	87.8	89.8	88.8	71.8	81.9	89.1	91.2	90.1	74.2	83.5	90.6	92.6	91.6	75.9
UniMatch	80.2	88.5	89.2	88.8	71.8	81.7	90.1	90.1	90.1	73.5	83.7	91.2	91.5	91.3	75.8	85.1	92.9	91.7	92.3	77.9
C2F-SemiCD	81.0	89.5	88.8	89.1	72.5	82.2	90.8	89.5	90.1	74.2	83.9	91.5	91.3	91.4	76.3	85.5	93.2	91.5	92.3	78.5
ISCDNet	81.5	89.2	**90.4**	89.8	73.1	82.6	89.7	**92.5**	91.1	74.6	83.7	91.6	90.7	91.1	76.0	85.2	91.7	**93.4**	92.5	78.8
PhysSANet	**82.3**	**90.4**	90.1	**90.2**	**73.8**	**83.1**	**91.8**	91.2	**91.5**	**75.3**	**84.8**	**92.2**	**91.8**	**92.0**	**76.9**	**86.2**	**93.9**	92.3	**93.1**	**79.5**
SupBaseline	54.1	76.8	70.2	73.3	62.1	60.9	78.5	76.1	77.3	64.8	68.4	82.9	80.5	81.7	67.5	75.9	86.0	86.1	86.0	70.5

TABLE V

COMPREHENSIVE PERFORMANCE COMPARISON ON SYSU-CD DATASET UNDER VARYING ANNOTATION RATIOS (%). BEST RESULTS ARE IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED. ALL VALUES REPRESENT MEAN OVER FIVE RUNS WITH DIFFERENT RANDOM SEEDS

Method	5%					10%					20%					40%				
	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ	IoU	P	R	F1	PQ
SupBaseline	60.0	74.0	76.0	75.0	46.0	61.3	75.1	76.9	76.0	47.3	62.6	76.2	77.8	77.0	48.6	64.0	77.3	78.8	78.0	49.9
S4GAN	55.7	72.5	70.5	71.5	40.5	57.0	73.5	71.5	72.5	42.0	58.3	74.5	72.8	73.6	43.5	59.7	75.6	74.0	74.8	44.7
SemiCDNet	57.1	72.5	73.0	72.7	42.0	58.4	73.6	73.9	73.7	43.5	59.7	74.7	74.8	74.7	45.0	61.1	75.8	75.8	75.8	46.3
SemiCD	57.6	68.4	78.5	73.1	43.0	58.9	69.2	79.8	74.1	44.3	60.2	70.0	81.2	75.2	45.6	61.6	70.8	82.5	76.2	47.0
FixMatch	59.8	75.2	72.8	74.0	46.2	61.1	76.8	74.5	75.6	47.8	62.4	78.5	75.9	77.2	48.9	63.4	79.2	76.1	77.6	49.2
UniMatch	62.2	77.0	76.4	76.7	49.0	63.5	78.0	77.3	77.6	50.2	64.8	79.0	78.3	78.6	51.5	66.2	80.1	79.3	79.7	52.7
ISCDNet	63.0	78.5	75.8	77.1	49.8	64.3	79.5	76.7	78.1	51.3	66.0	80.8	78.0	79.1	52.6	67.7	82.1	79.5	80.2	53.8
PhysSANet	**63.6**	**80.0**	**75.1**	**77.5**	**50.5**	**64.9**	**81.2**	**76.0**	**78.5**	**52.0**	**66.8**	**82.5**	**78.5**	**79.6**	**53.5**	**68.5**	**83.6**	**80.2**	**80.7**	**54.6**

TABLE VI

COMPUTATIONAL COMPLEXITY AND EFFICIENCY ANALYSIS. TRAINING TIME IS NORMALIZED RELATIVE TO SEMICDNET. INFERENCE LATENCY IS MEASURED ON 256 × 256 IMAGE PAIRS

Method	Params (M)	FLOPs (G)	Train Time	Inference (ms)	Memory (GB)
SemiCDNet	52.40	153.00	1.00×	42.3	5.2
S4GAN	49.62	148.23	1.68×	45.7	5.8
SemiCD	50.70	151.80	1.85×	48.2	5.5
FixMatch	38.25	145.60	1.52×	38.5	4.8
UniMatch	40.47	163.89	1.74×	41.8	5.3
C2F-SemiCD	46.32	168.45	1.92×	49.1	6.1
ISCDNet	51.65	174.23	1.89×	52.3	6.5
PhysSANet	**48.85**	**182.20**	**1.84×**	**51.6**	**6.8**

D. Comparative Performance Analysis

We compare PhysSANet against several state-of-the-art methods across three categories: 1) *general SSL methods*, FixMatch [28] and UniMatch [29], which are strong domain-agnostic baselines widely validated in various vision tasks; 2) *semi-supervised CD methods*, SemiCD [30], S4GAN [31], SemiCDNet [32], C2F-SemiCD [22], and ISCDNet [19], which incorporate remote-sensing-specific adaptations; and 3) a *supervised baseline* (SupBaseline) trained only on the labeled subset without exploiting unlabeled data. All methods are evaluated under identical experimental settings across four annotation ratios (5%, 10%, 20%, and 40%) on three benchmark datasets, ensuring a fair and comprehensive comparison.

1) Overall Performance Comparison: We evaluate PhysSANet on WHU-CD, LEVIR-CD, and SYSU-CD under varying annotation ratios (5%, 10%, 20%, and 40%). Our method consistently outperforms recent semi-supervised baselines across all configurations. Tables III–V report five complementary metrics (IoU, precision, recall, $F1$-score, and PQ). Statistical significance is assessed via paired t-tests ($p < 0.05$) over five independent runs.

Cross-Dataset Robustness Under Low-Data Regimes: PhysSANet achieves the best IoU on all three datasets, with the most notable advantages under scarce annotation. At 5%

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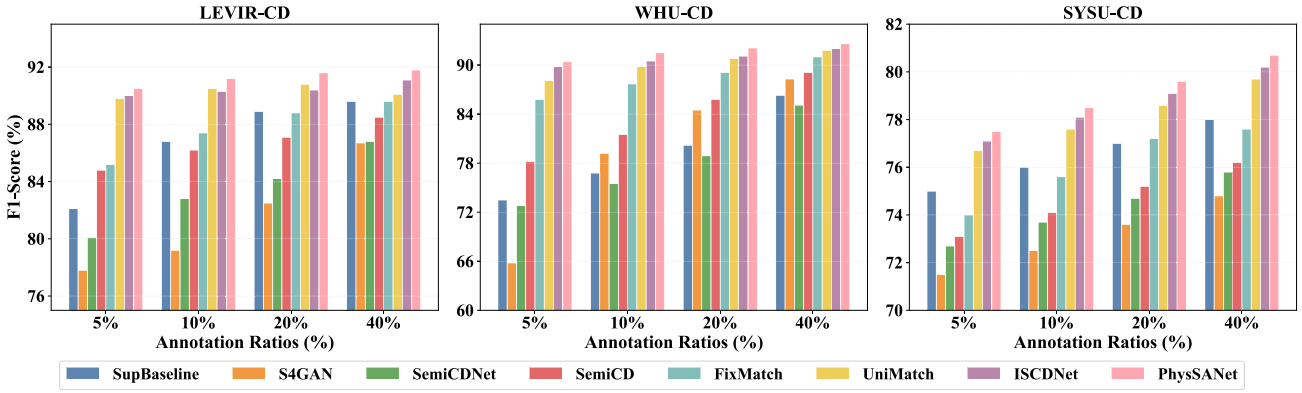


Fig. 5. $F1$-score comparison across three benchmark datasets under different annotation ratios. Each bar group corresponds to a specific supervision level (5%, 10%, 20%, and 40%), and each color denotes a competing method. PhysSANet consistently achieves the highest performance across all annotation budgets, highlighting its strong label efficiency and superior generalization ability in low-label regimes.

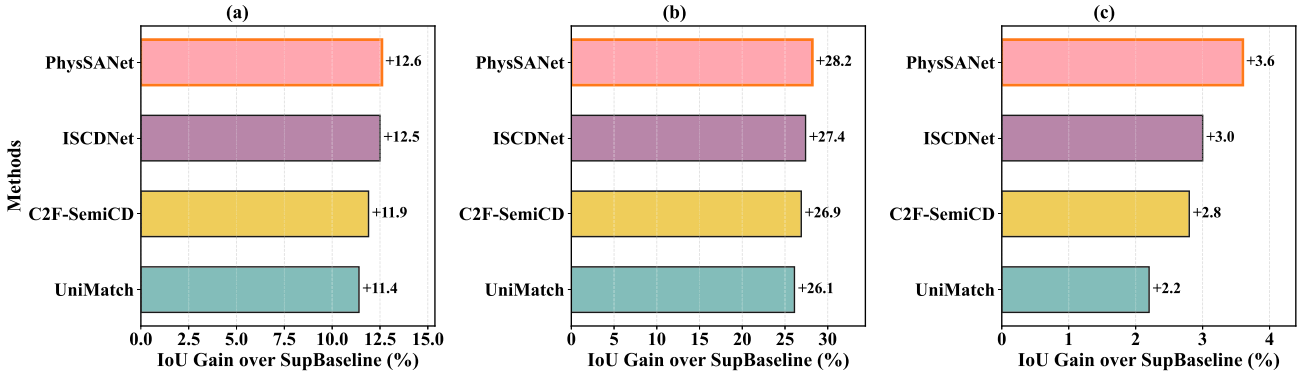


Fig. 6. IoU performance gain over supervised baseline at 5% annotation ratio on (a) LEVIR-CD, (b) WHU-CD, and (c) SYSU-CD datasets. Each bar represents the IoU improvement of a semi-supervised method relative to the fully supervised baseline. PhysSANet (bottom bar with thick orange border) achieves the largest gains across all three datasets, demonstrating superior capability in exploiting unlabeled data.

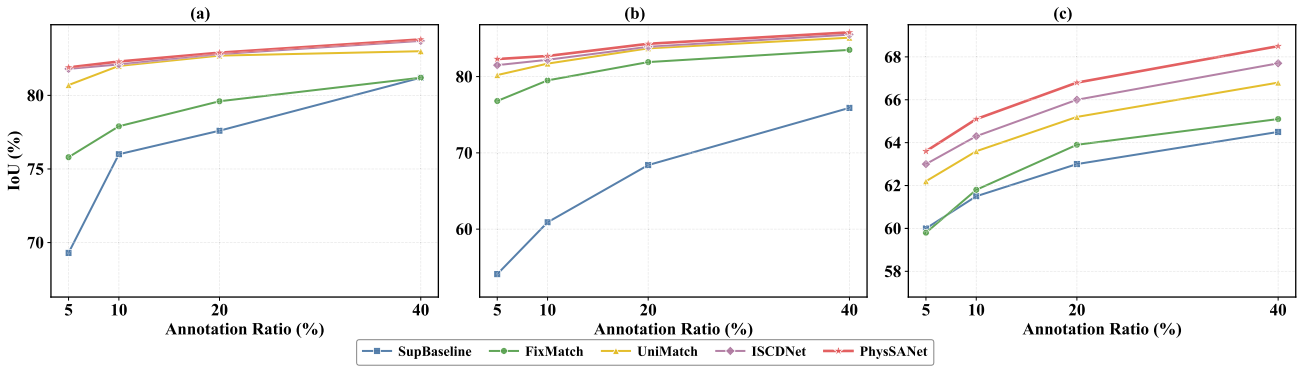


Fig. 7. Annotation efficiency analysis showing IoU versus annotation ratio for five representative methods. (a) LEVIR-CD, (b) WHU-CD, and (c) SYSU-CD datasets. PhysSANet (pink line with star markers) consistently outperforms all competing methods across all annotation levels. The curves demonstrate that PhysSANet achieves superior data efficiency, requiring significantly fewer labeled samples to reach equivalent or better performance than other approaches.

labels, our method reaches 82.3% IoU on WHU-CD, 81.9% on LEVIR-CD, and 63.6% on SYSU-CD. Compared with C2F-SemiCD, PhysSANet improves IoU by +1.3, +0.7, and +0.8 percentage points (pps), respectively; compared with ISCDNet, the gains are +0.8, +0.1, and +1.4 pp. These improvements hold on both building-focused datasets (WHU-CD and LEVIR-CD) and the more heterogeneous SYSU-CD, indicating good cross-dataset robustness of the proposed physics-inspired semi-supervised framework. Relative to the general-purpose UniMatch baseline (without remote-sensing-specific priors), PhysSANet improves IoU by +2.1, +1.2, and +1.4 pp at 5% labels, suggesting the benefit of integrating acquisition-aligned perturbations and structure-aware constraints for remote sensing CD.

Balanced Multimetric Performance and Scalability to Higher Supervision: PhysSANet maintains favorable precision–recall tradeoffs while achieving consistently high $F1$-scores across all annotation ratios. Precision ranges from

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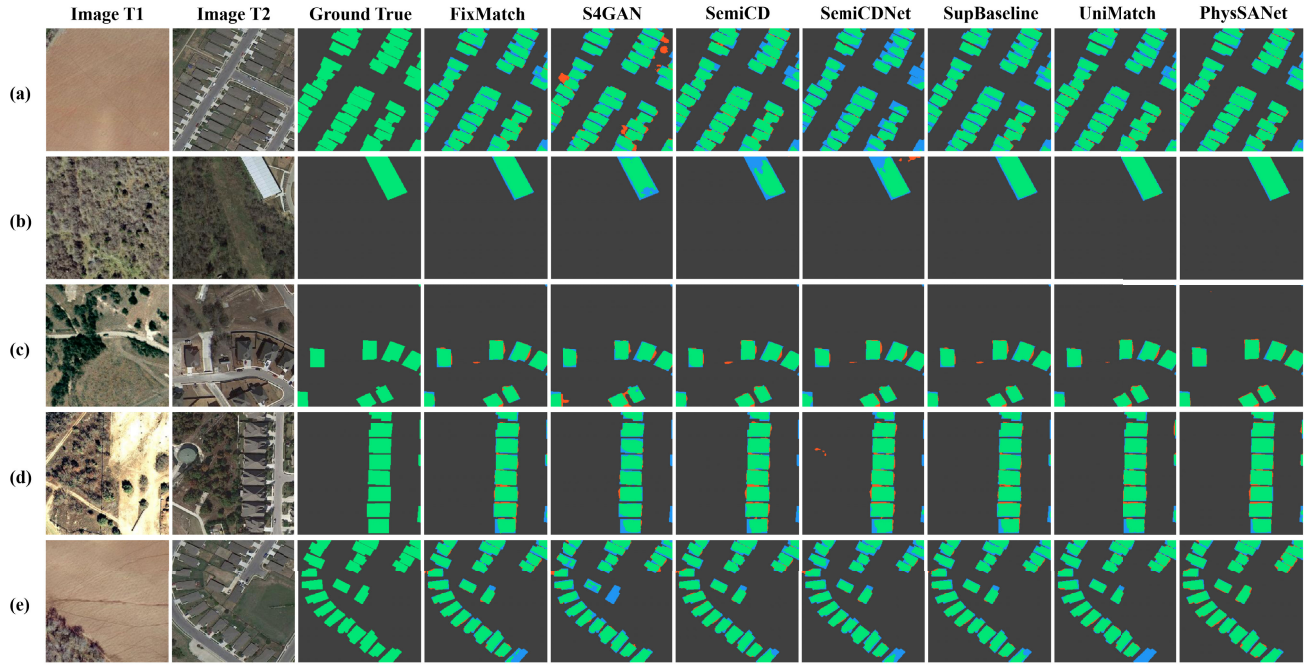


Fig. 8. Qualitative error analysis on the LEVIR-CD dataset under 20% annotation. Color coding follows Fig. 9. The results highlight our method's effectiveness in building CD with improved structural consistency and fewer boundary artifacts. The part labels (a)–(e) denote five representative test examples.

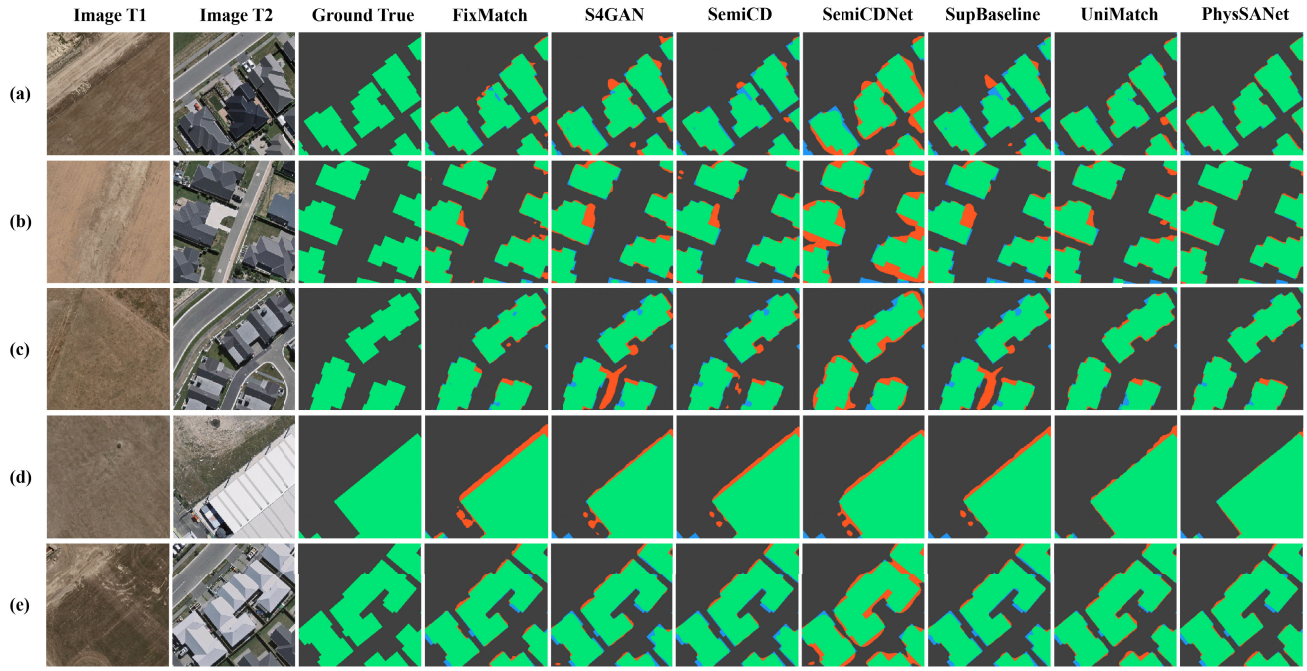


Fig. 9. Qualitative error analysis on the WHU-CD dataset under 40% annotation. Color coding: gray = TNs, green = TPs, red = FP, and blue = FNs. PhysSANet demonstrates boundary preservation and reduced false detections compared to baseline methods. The part labels (a)–(e) denote five representative test examples.

80.0% to 93.9% across datasets and label ratios, while recall spans 75.1%–92.3%, which is consistent with the proposed confidence–plausibility pseudo-label filtering that balances detection completeness and reliability. Notably, gains remain visible even at higher annotation ratios (40%), indicating complementary benefits beyond simply increasing labeled data. At 40% labels, PhysSANet still outperforms C2F-SemiCD by +0.7, +0.2, and +0.8 pp IoU on WHU-CD, LEVIR-CD, and SYSU-CD, respectively. The PQ metric, which emphasizes instance-level geometric fidelity, also improves notably on building-centric datasets (up to 79.9% on WHU-CD), supporting that the structure-aware components help preserve geometric regularities essential for accurate change localization.

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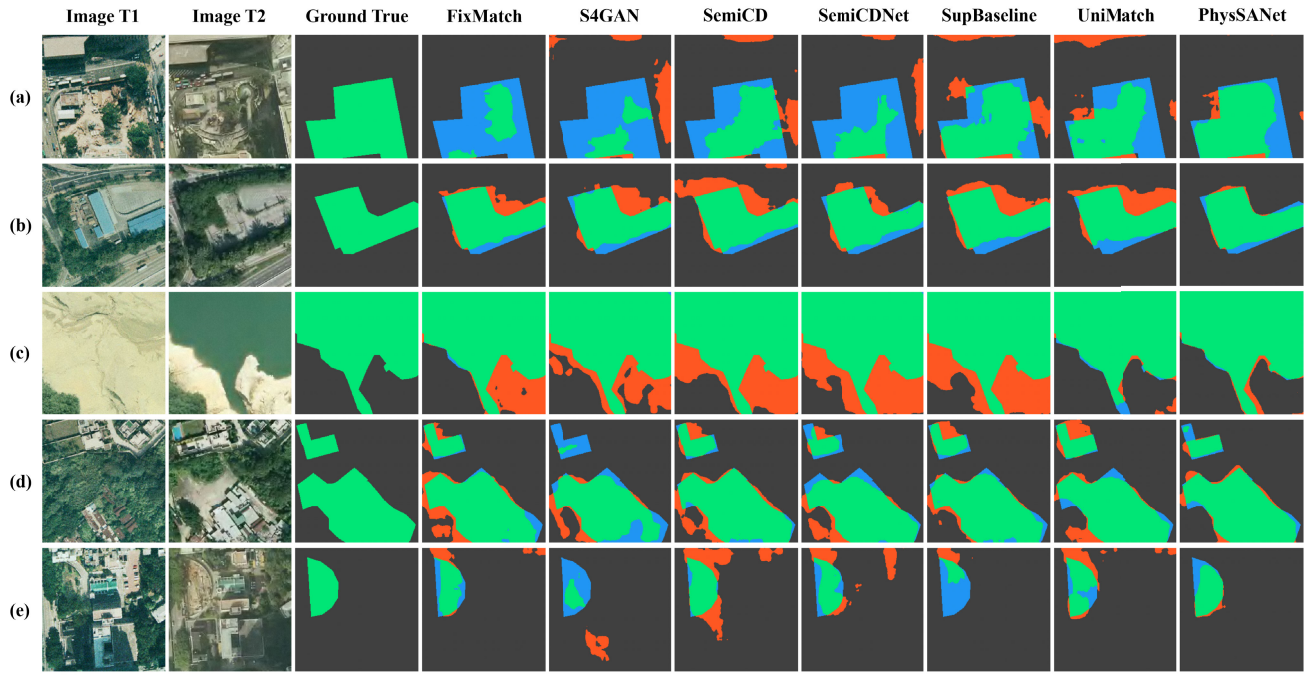


Fig. 10. Qualitative error analysis on the SYSU-CD dataset under 5% annotation. Color coding follows Fig. 9. Under extreme label scarcity, PhysSANet maintains robust performance, thanks to physics-inspired perturbation (augmentation) and structure-aware attention. The part labels (a)–(e) denote five representative test examples.

2) Computational Efficiency Analysis: TABLE VI summarizes the computational complexity and efficiency of representative methods on our experimental hardware (two NVIDIA RTX 4090 GPUs). PhysSANet achieves a practical balance between accuracy and efficiency, requiring 48.85M parameters and 182.20 GFLOPs per forward pass on 256 × 256 image pairs. While this represents a moderate parameter increase over UniMatch (40.47M), the additional cost is accompanied by consistent IoU gains across datasets and annotation ratios.

Training time for PhysSANet is approximately 1.84× that of SemiCDNet and is comparable to other recent methods such as C2F-SemiCD (1.92×) and ISCDNet (1.89×). The additional cost mainly comes from PAP generation and MHSAA, which improve pseudo-label quality and boundary accuracy. Inference latency is 51.6 ms per 256 × 256 image pair [about 19 frames per second (FPS)], which is sufficient for many time-sensitive monitoring scenarios (e.g., rapid damage assessment). Peak GPU memory usage is 6.8 GB during training, fitting comfortably within modern GPU memory budgets.

Overall, PhysSANet introduces modest overhead while delivering higher accuracy, yielding a favorable tradeoff. The modular design also enables flexible deployment; for example, disabling MHSAA reduces the parameter count by about 15% while retaining most of the performance gains, which can benefit resource-constrained settings.

E. Visualization and Qualitative Analysis

1) Multimetric Performance Comparison: Fig. 5 summarizes $F1$-scores across different annotation ratios. PhysSANet consistently outperforms all baselines on the three datasets, and the performance margins increase under low-label regimes, indicating strong label efficiency. Fig. 6 further compares IoU gains over the *supervised* model trained with the same annotation ratio (5%). Here, "+x%" denotes the *relative* improvement, i.e., $(\text{IoU}_{\text{method}} - \text{IoU}_{\text{sup}})/\text{IoU}_{\text{sup}} \times 100\%$. The horizontal bar chart shows that PhysSANet achieves the largest relative IoU gains across all three datasets (WHU-CD: +28.2%, LEVIR-CD: +12.6%, and SYSU-CD: +3.6%), highlighting its stronger ability to exploit unlabeled data than prior semi-supervised methods. Fig. 7 reports annotation–efficiency curves. On WHU-CD and LEVIR-CD, PhysSANet with 5% labels surpasses the supervised baseline with 40% labels, corresponding to approximately 87.5% annotation savings. Overall, these visualizations collectively demonstrate the robustness and practical value of PhysSANet for semi-supervised CD across diverse scenarios.

2) Qualitative Improvements: Figs. 8–10 visualize error maps using a consistent true positive (TP)/true negative (TN)/FP/false negative (FN) color scheme. Three advantages are observed.

1) *Boundary Precision:* PhysSANet produces sharper and more regular building outlines, reducing jagged or fragmented predictions in challenging areas. This is consistent with the role of SSA-FPN and MHSAA in preserving geometric structure during multiscale fusion.
2) *Robustness to Illumination and Shadows:* PhysSANet yields fewer FPs in shadowed or poorly lit regions, suggesting that PAPs help suppress illumination-induced artifacts and better distinguish genuine changes from photometric variations.
3) *Sensitivity to Small and Heterogeneous Changes:* On the heterogeneous SYSU-CD dataset, PhysSANet detects small or elongated changes while reducing false alarms, decreasing both FN and FP. This behavior is consistent with the structure-aware attention design, which enhances discrimination of subtle temporal differences.

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TABLE VII

ABLATION STUDY OF PHYSSANET COMPONENTS ON THREE DATASETS AT 40% ANNOTATION RATIO. EACH MODULE IS EVALUATED INDIVIDUALLY AND IN COMBINATION. BEST RESULTS ARE SHOWN IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Configuration	PAP	MHSAA	SSA-FPN	MVCR	LEVIR-CD			WHU-CD			SYSU-CD		
					IoU	F1	PQ	IoU	F1	PQ	IoU	F1	PQ
Baseline and Single Module Ablations													
Baseline	✗	✗	✗	✗	81.2	89.6	72.8	75.9	86.0	70.5	64.0	78.0	49.9
+ PAP (Full)	✓	✗	✗	✗	82.5	90.6	74.2	79.1	88.2	73.1	65.2	79.0	51.7
+ MHSAA	✗	✓	✗	✗	82.2	90.3	73.9	78.5	87.9	72.5	64.8	78.7	51.4
+ SSA-FPN	✗	✗	✓	✗	82.1	90.5	74.2	80.2	89.1	74.2	65.5	79.2	52.1
+ MVCR	✗	✗	✗	✓	81.8	90.1	73.5	78.1	87.5	72.1	64.5	78.4	51.1
PAP Sub-component Ablations													
+ PAP (Veiling only)	$✓^{v}$	✗	✗	✗	81.8	90.2	73.5	77.2	87.8	71.8	64.6	78.5	50.8
+ PAP (Illumination only)	$✓^{l}$	✗	✗	✗	82.1	90.4	74.0	78.5	88.0	72.6	65.0	78.8	51.3
+ PAP (Noise only)	$✓^{n}$	✗	✗	✗	81.5	89.9	73.2	76.8	87.3	71.2	64.3	78.3	50.5
+ PAP (Veiling + Illumination)	$✓^{vl}$	✗	✗	✗	82.3	90.5	74.1	78.8	88.1	73.0	65.1	78.9	51.5
+ PAP (Veiling + Noise)	$✓^{vn}$	✗	✗	✗	81.9	90.3	73.7	77.5	87.9	72.0	64.7	78.6	51.0
+ PAP (Illumination + Noise)	$✓^{ln}$	✗	✗	✗	82.2	90.4	73.9	78.6	88.0	72.7	64.9	78.7	51.2
+ PAP (Full: All)	$✓^{vln}$	✗	✗	✗	82.5	90.6	74.2	79.1	88.2	73.1	65.2	79.0	51.7
MHSAA Branch Ablations													
+ MHSAA (Edge only)	✗	$✓^{e}$	✗	✗	81.8	90.2	73.6	77.5	87.6	71.9	64.5	78.4	51.0
+ MHSAA (Struct only)	✗	$✓^{s}$	✗	✗	81.9	90.2	73.7	77.8	87.8	72.2	64.6	78.5	51.1
+ MHSAA (Cross only)	✗	$✓^{t}$	✗	✗	81.7	90.1	73.5	77.3	87.5	71.7	64.4	78.3	50.9
+ MHSAA (Edge + Struct)	✗	$✓^{es}$	✗	✗	82.0	90.3	73.8	78.1	87.8	72.4	64.7	78.6	51.3
+ MHSAA (Full: All)	✗	$✓^{est}$	✗	✗	82.2	90.3	73.9	78.5	87.9	72.5	64.8	78.7	51.4
Two-Module Combinations													
PAP + MHSAA	✓	✓	✗	✗	82.9	91.0	74.8	81.8	90.1	75.5	65.8	79.5	52.3
PAP + SSA-FPN	✓	✗	✓	✗	82.8	90.9	75.5	83.1	91.2	76.8	66.5	80.1	53.2
PAP + MVCR	✓	✗	✗	✓	82.7	90.7	74.5	81.2	89.8	74.8	65.5	79.3	52.0
MHSAA + SSA-FPN	✗	✓	✓	✗	82.5	90.8	75.2	82.5	90.8	76.2	66.3	79.9	53.0
MHSAA + MVCR	✗	✓	✗	✓	82.4	90.5	74.2	80.8	89.5	74.0	65.1	79.0	51.8
SSA-FPN + MVCR	✗	✗	✓	✓	82.3	90.6	74.8	81.5	90.2	75.5	65.8	79.6	52.5
Three-Module Combinations													
PAP + MHSAA + SSA-FPN	✓	✓	✓	✗	<u>83.5</u>	<u>91.5</u>	<u>76.2</u>	<u>85.5</u>	<u>92.5</u>	<u>79.2</u>	<u>67.2</u>	<u>80.5</u>	<u>54.2</u>
PAP + MHSAA + MVCR	✓	✓	✗	✓	83.1	91.2	75.5	83.8	91.8	77.5	66.5	80.0	53.5
PAP + SSA-FPN + MVCR	✓	✗	✓	✓	83.2	91.3	76.0	84.8	92.1	78.5	66.8	80.2	53.8
MHSAA + SSA-FPN + MVCR	✗	✓	✓	✓	82.8	91.0	75.8	83.2	91.2	77.0	66.5	80.0	53.2
PhysSANet (Full)	✓	✓	✓	✓	**83.9**	**91.8**	**76.6**	**86.2**	**93.1**	**79.9**	**67.6**	**80.7**	**54.6**

^{v}Haze-like veiling perturbation; ^{l}Illumination-related perturbation; ^{n}Sensor noise.
^{e}Edge-aware attention; ^{s}Structure-aware attention; ^{t}Cross-temporal attention.

We conduct comprehensive ablation studies on all three datasets at a 40% annotation ratio to systematically quantify the contribution of each component. All variants share identical training protocols and hyperparameters, with results averaged over five runs to ensure statistical reliability (Table VII).

3) Individual Component Analysis: All four modules consistently improve performance over the baseline across diverse scenarios. PAP yields the largest gains (+1.3% IoU on LEVIR-CD, +3.2% on WHU-CD, and +1.2% on SYSU-CD), indicating that acquisition-aligned perturbations provide more reliable consistency signals than generic natural-image augmentations. MHSAA and SSA-FPN further enhance performance through explicit geometric modeling and multiscale boundary preservation, while MVCR refines pseudo-labels via appearance-plausibility-aware consistency regularization. The pronounced gains on WHU-CD reflect its sensitivity to geometric fidelity and illumination-related appearance variability.

4) PAP Component Decomposition: Fig. 11 analyzes the contribution of different appearance-variation components within PAP.

1) *Illumination-related perturbations* contribute the largest gains among single components, with +0.9% IoU on LEVIR-CD and + 2.6% on WHU-CD (Table VII), indicating that modeling illumination/exposure differences is particularly important for building CD where radiometric variations can be comparable to true change signals.
2) *Haze-like veiling perturbations* provide consistent improvements (+0.6% on LEVIR-CD and +1.3% on WHU-CD), demonstrating robustness to visibility-related degradation commonly observed in multitemporal imagery.
3) *Sensor noise perturbations* yield smaller but nonnegligible gains (+0.3% on LEVIR-CD and +0.9% on WHU-CD) and further improve performance when combined with other components.

Pairwise combinations consistently outperform individual components, e.g., veiling plus illumination achieves 82.3% IoU on LEVIR-CD, higher than either veiling-only (81.8%) or illumination-only (82.1%). The full PAP configuration reaches 82.5% IoU, suggesting that jointly modeling multiple acquisition-related appearance variations provides the

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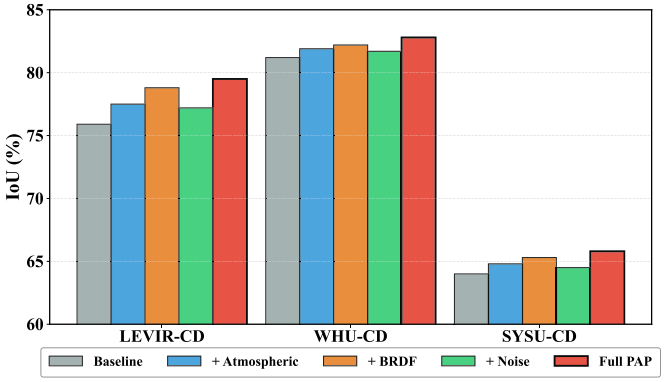


Fig. 11. Contribution analysis of PAP subcomponents at 40% annotation ratio. Each bar group shows the baseline (gray), individual appearance-variation components (blue, orange, and green), and the full PAP configuration (red). Illumination-related perturbations account for approximately 45%–50% of the total performance gains, followed by haze-like veiling and sensor noise. The full PAP configuration consistently achieves super-additive improvements (+1.3%, +3.2%, and +1.2% IoU on LEVIR-CD, WHU-CD, and SYSU-CD, respectively), demonstrating the benefit of jointly modeling multiple acquisition-related appearance variations.

most reliable perturbations for semi-supervised consistency learning.

5) Structure-Aware Attention Decomposition: The third block of Table VII analyzes the attention branches within MHSAA. Each branch targets complementary geometric cues and yields distinct benefits. Edge-aware attention improves boundary localization (+0.6% IoU), structure-aware attention enhances region-level coherence (+0.7% IoU), and cross-temporal attention suppresses false detections caused by appearance inconsistencies (+0.5% IoU). The full MHSAA module achieves 82.2% IoU on LEVIR-CD, outperforming all individual branches and confirming that explicit geometric cues are complementary when combined through adaptive weighting.

6) Module Synergy and Complete Framework: Two- and three-module combinations demonstrate consistent complementary benefits across datasets. For example, PAP combined with SSA-FPN improves LEVIR-CD IoU to 82.8%, outperforming PAP-only (82.5%) and SSA-FPN-only (82.1%), highlighting the compatibility between acquisition-aligned perturbations and structure-preserving multiscale fusion. The three-module configuration PAP + MHSAA + SSA-FPN further improves performance to 83.5% IoU by jointly enhancing appearance robustness, geometric attention, and boundary-aware fusion. Incorporating MVCR yields the full PhysSANet framework with 83.9% IoU on LEVIR-CD and the best overall results on WHU-CD and SYSU-CD, where appearance-plausibility-aware pseudo-label refinement is particularly beneficial under complex acquisition conditions.

7) Computational Efficiency: Table VIII summarizes the per-module computational overhead. PAP introduces no inference cost, as it is used only during training, whereas MHSAA and SSA-FPN contribute 28.3 GFLOPs (15.5%) and 45.1 GFLOPs (24.7%), respectively. Despite the increase from 108.8 to 182.2 GFLOPs, PhysSANet maintains 51.6 ms per image pair (19.4 FPS) on 256 $\times$ 256 inputs, which is only about 21% slower than the baseline latency of 42.3 ms (23.6 FPS). Given the substantial accuracy improvements (+2.7%–10.3% IoU across datasets), this overhead remains acceptable for practical remote sensing CD applications.

TABLE VIII
PER-MODULE COMPUTATIONAL COST

Component	GFLOPs	% of Total
Backbone (ResNet-50)	108.8	59.7%
MHSAA	28.3	15.5%
SSA-FPN	45.1	24.7%
Prediction Head	0.0	0.1%
Total	**182.2**	**100%**

TABLE IX
SENSITIVITY ANALYSIS OF THE STRONG-VIEW PAP STRENGTH α_{STRONG}, THE APPEARANCE-PLAUSIBILITY THRESHOLD τ_p, AND THE CONFIDENCE THRESHOLD τ_c UNDER THE 5% LABELSETTING ON LEVIR-CD AND WHU-CD. IOU AND $F1$ DENOTE THE MEAN OVER FIVE RUNS, AND STD DENOTES THE STANDARD DEVIATION OF IOU ACROSS RUNS

Parameter	Value	LEVIR-CD (5%)			WHU-CD (5%)		
		IoU	F1	Std	IoU	F1	Std
α_{strong}	0.3	81.1	89.7	0.41	81.4	89.4	0.44
	0.5	81.5	90.1	0.33	81.8	89.8	0.36
	0.6	81.7	90.3	0.30	82.1	90.0	0.33
	0.7	81.6	90.2	0.31	81.9	89.9	0.34
	0.9	80.8	89.4	0.45	81.0	89.1	0.48
	[0.5, 0.8]	**81.9**	**90.5**	**0.28**	**82.3**	**90.2**	**0.31**
τ_p	0.60	81.2	89.8	0.40	81.5	89.5	0.43
	0.70	81.7	90.3	0.34	82.0	89.9	0.35
	0.80	**81.9**	**90.5**	**0.28**	**82.3**	**90.2**	**0.31**
	0.90	81.6	90.2	0.32	82.1	90.0	0.33
	0.95	81.3	89.9	0.38	81.6	89.6	0.40
τ_c	0.80	81.1	89.7	0.42	81.4	89.4	0.45
	0.85	81.4	90.0	0.37	81.8	89.8	0.38
	0.90	81.7	90.3	0.31	82.1	90.1	0.33
	0.95	**81.9**	**90.5**	**0.28**	**82.3**	**90.2**	**0.31**
	0.99	81.3	89.9	0.39	81.6	89.6	0.41

8) Sensitivity Analysis of PAP: To further evaluate the robustness of PhysSANet to key hyperparameter choices, we analyze the sensitivity of three representative parameters: the strong-view PAP strength α_{strong}, the appearance-plausibility threshold τ_p, and the confidence threshold τ_c. All analyses are conducted under the 5% label setting on LEVIR-CD and WHU-CD, where pseudo-label noise and consistency instability are more likely to occur, making this setting particularly informative for evaluating hyperparameter robustness. In each experiment, only the target hyperparameter is varied while all remaining settings are fixed to their default values. The results are summarized in Table IX.

For α_{strong}, performance improves as the perturbation strength increases from 0.3 to a moderate range (0.5–0.7), but degrades when the perturbation becomes excessively strong (0.9). This is consistent with the design motivation of PAP: weak perturbations provide insufficient acquisition-aligned variability, whereas overly strong perturbations corrupt structural correspondence and increase pseudo-label noise. The default interval-sampling strategy $[0.5, 0.8]$ consistently outperforms all fixed values on both datasets while achieving lower run-to-run variance, suggesting that stochastic strength variation within a moderate range provides a more stable

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training signal than any single fixed value. The performance advantage of interval sampling over any single fixed value (e.g., 81.9% versus 81.7% IoU at the best fixed point $\alpha_{\text{strong}} = 0.6$ on LEVIR-CD) can be attributed to the additional training diversity introduced by stochastic strength variation, which exposes the model to a broader range of acquisition-aligned appearance conditions within a single training run.

For τ_p, moderate threshold values yield the most favorable tradeoff between pseudo-label quality and supervision coverage. When τ_p is too low (0.60), the plausibility filter admits noisier pseudo-labels; when it is too high (0.95), filtering becomes overly conservative and reduces usable supervision. The default $\tau_p = 0.80$ achieves the best results on both datasets. For τ_c, a similar but slightly sharper trend is observed. When τ_c is too low (0.80), low-confidence pseudo-labels are retained and introduce additional noise into consistency learning. When it is too high (0.99), the retained supervision becomes overly sparse. The default $\tau_c = 0.95$ provides the most favorable balance between pseudo-label coverage and noise suppression on both datasets.

F. Discussion

1) Key Findings: Our experiments provide three main insights for semi-supervised remote sensing CD. First, acquisition-aligned perturbations are more effective than generic transformations because they expose the model to realistic appearance variability commonly observed in remote sensing imagery. In particular, illumination-related perturbations are the most influential single factor within PAP, and combining veiling, illumination, and sensor noise yields larger gains than any individual component, suggesting that jointly modeling multiple acquisition-related variations provides stronger consistency supervision than using a single factor alone. Second, explicit geometric modeling is important: structure-aware attention and coordinate-aware multiscale fusion consistently improve boundary delineation and region coherence on building-focused datasets, where thin structures and sharp object edges are critical. Third, hierarchical multiview consistency helps suppress pseudo-label noise beyond confidence thresholding alone and is associated with more stable training under radiometric discrepancies such as haze-like veiling and illumination shifts.

The larger gains on building-centric datasets (e.g., LEVIR-CD and WHU-CD) than on more heterogeneous scenes (e.g., SYSU-CD) indicate that acquisition-aligned SSL is particularly effective when geometric regularity dominates. A plausible explanation is twofold. On the one hand, the perturbations modeled by PAP, including atmospheric veiling, illumination/exposure shifts, and sensor noise, primarily affect radiometric appearance, while the geometric boundaries of stable man-made structures remain largely invariant; encouraging invariance to such effects, therefore, directly benefits building CD. On the other hand, dominant change patterns in SYSU-CD, such as vegetation phenology and water-body variation, are driven more by intrinsic spectral–temporal dynamics than by acquisition-induced radiometric drift, and are, therefore, less directly addressed by the acquisition-aligned perturbations modeled by PAP. This suggests a broader design principle: domain-specific perturbations should emphasize invariances that are most relevant to the target task, and for man-made structures, these invariances are more closely related to illumination and acquisition-related appearance shifts than to arbitrary color or geometric distortions. This interpretation is also consistent with the PAP-strength sensitivity analysis (Table IX), which shows that performance degrades at both extremes of α_{strong} and that the default interval-sampling strategy provides the most favorable balance between perturbation diversity and training stability.

2) Limitations: The current perturbation design focuses on common acquisition-related appearance variations and bitemporal image pairs, which may be less effective for gradual or cyclic changes (e.g., vegetation phenology) and for long multitemporal sequences. The framework currently addresses binary CD only; extending it to semantic CD may require class-aware regularization and more fine-grained priors on region consistency. Moreover, although plausibility-aware pseudo-label filtering improves robustness in the studied setting, sensitivity to severe domain shifts (e.g., cross-sensor generalization or drastic resolution changes) remains an open challenge. Although the current sensitivity analysis covers the key PAP-related hyperparameters, a broader characterization of additional weighting coefficients remains for future work.

3) Future Directions: Promising extensions include temporal consistency modeling for recurrent changes, cross-modal fusion with synthetic aperture radar (SAR) to leverage complementary scattering cues, and integration with large-scale pretrained models to further improve transferability under domain shifts.

V. Conclusion

This article presented PhysSANet, a semi-supervised framework for remote sensing CD that couples acquisition-aligned perturbations with structure-aware representation learning. By modeling typical acquisition-related appearance variations in the augmentation pipeline and combining them with MHSAA and hierarchical MVCR, PhysSANet improves robustness and label efficiency under realistic acquisition variability.

The experimental results on three benchmarks show that, under limited supervision, PhysSANet consistently outperforms strong semi-supervised baselines while maintaining practical efficiency (48.85M parameters and 51.6-ms inference time). Ablation studies further demonstrate that PAP, the structure-aware modules, and the MVCR contribute complementary gains, with their combination yielding the best overall performance. These results indicate that incorporating acquisition-aligned perturbations into structure-aware SSL is an effective direction for robust remote sensing CD. Future work will extend this framework to multitemporal monitoring, cross-modal fusion, and the integration of foundation models for broader Earth observation applications.

References

[1] L. Wang, M. Zhang, X. Gao, and W. Shi, "Advances and challenges in deep learning-based change detection for remote sensing images: A review through various learning paradigms," *Remote Sens.*, vol. 16, no. 5, p. 804, Feb. 2024.

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[2] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[3] Q. Zhu, X. Guo, Z. Li, and D. Li, "A review of multi-class change detection for satellite remote sensing imagery," *Geo-Spatial Inf. Sci.*, vol. 27, no. 1, pp. 1–15, Jan. 2024.

[4] H. Gupta, S. Sil, A. Gangopadhyay, and G. Gawarkiewicz, "Observed surface and subsurface marine heat waves in the bay of Bengal from in-situ and high-resolution satellite data," *Climate Dyn.*, vol. 62, no. 1, pp. 203–221, Jan. 2024.

[5] M. A. Zanella, R. N. Martins, F. M. da Silva, L. C. C. Carvalho, M. de Carvalho Alves, and J. T. F. Rosas, "Coffee yield prediction using high-resolution satellite imagery and crop nutritional status in Southeast Brazil," *Remote Sens. Appl., Soc. Environ.*, vol. 33, Jan. 2024, Art. no. 101092.

[6] Y. Li, B. Dang, W. Li, and Y. Zhang, "GLH-water: A large-scale dataset for global surface water detection in large-size very-high-resolution satellite imagery," in *Proc. AAAI Conf. Artif. Intell.*, 2024, vol. 38, no. 20, pp. 22213–22221.

[7] E. T. Wasehun, L. Hashemi Beni, and C. A. Di Vittorio, "UAV and satellite remote sensing for inland water quality assessments: A literature review," *Environ. Monitor. Assessment*, vol. 196, no. 3, p. 277, Mar. 2024.

[8] W. Jiang, Y. Sun, L. Lei, G. Kuang, and K. Ji, "Change detection of multisource remote sensing images: A review," *Int. J. Digit. Earth*, vol. 17, no. 1, Dec. 2024, Art. no. 2398051.

[9] R. Jin, Z. Jia, X. Yin, Y. Niu, and Y. Qi, "Domain feature decomposition for efficient object detection in aerial images," *Remote Sens.*, vol. 16, no. 9, p. 1626, May 2024.

[10] J. Chen, D. Hou, C. He, Y. Liu, Y. Guo, and B. Yang, "Change detection with cross-domain remote sensing images: A systematic review," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 11563–11582, 2024.

[11] M. He, J. Zhang, Y. He, X. Zuo, and Z. Gao, "Annotated dataset for training cloud segmentation neural networks using high-resolution satellite remote sensing imagery," *Remote Sens.*, vol. 16, no. 19, p. 3682, Oct. 2024.

[12] X. Zhang et al., "Remote sensing image semantic change detection boosted by semi-supervised contrastive learning of semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5624113.

[13] H. Chen, T. Xu, Z. Chen, P. Liu, H. Bai, and J. Li, "Multi-scale change-aware transformer for remote sensing image change detection," in *Proc. 32nd ACM Int. Conf. Multimedia*, Oct. 2024, pp. 2992–3000.

[14] J. Su, X. Yu, X. Wang, Z. Wang, and G. Chao, "Enhanced transfer learning with data augmentation," *Eng. Appl. Artif. Intell.*, vol. 129, Mar. 2024, Art. no. 107602.

[15] J.-X. Wang, T. Li, S.-B. Chen, J. Tang, B. Luo, and R. C. Wilson, "Reliable contrastive learning for semi-supervised change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4416413.

[16] L. Hu, S. Li, J. Ruan, and F. Gao, "SemiPSENet: A novel semi-supervised change detection network for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5634214.

[17] L. Ding et al., "A survey of sample-efficient deep learning for change detection in remote sensing: Tasks, strategies, and challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 13, no. 3, pp. 164–189, Sep. 2025.

[18] M. Cai, H. Chen, T. Zhang, Y. Zhuang, and L. Chen, "Consistency regularization based on masked image modeling for semisupervised remote sensing semantic segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17442–17460, 2024.

[19] Y. Zuo et al., "Robust instance-based semi-supervised learning change detection for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4404815.

[20] W. M. Shafin, T. A. Joty, and S. M. M. Ahsan, "NSNUNet: A neighbourhood attention-based Siamese nested unet for change detection using satellite images," in *Proc. 27th Int. Conf. Comput. Inf. Technol. (ICCIT)*, Dec. 2024, pp. 2718–2723.

[21] Y. Watanabe and T. Kamiya, "Detection of temporal changes from satellite images based on improved SNUNet-CD," in *Proc. Joint 13th Int. Conf. Soft Comput. Intell. Syst. 25th Int. Symp. Adv. Intell. Syst. (SCIS&ISIS)*, Nov. 2024, pp. 1–4.

[22] C. Han, C. Wu, M. Hu, J. Li, and H. Chen, "C2F-SemiCD: A coarse-to-fine semi-supervised change detection method based on consistency regularization in high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4702621.

[23] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[24] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8007805.

[25] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[26] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[27] A. Tarvainen and H. Valpola, "Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results," in *Proc. Int. Conf. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1195–1204.

[28] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.

[29] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.

[30] W. Gedara Chaminda Bandara and V. M. Patel, "Revisiting consistency regularization for semi-supervised change detection in remote sensing images," 2022, *arXiv:2204.08454*.

[31] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.

[32] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[33] Q. Gui, H. Zhou, N. Guo, and B. Niu, "A survey of class-imbalanced semi-supervised learning," *Mach. Learn.*, vol. 113, no. 8, pp. 5057–5086, Aug. 2024.

[34] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2613–2622.

[35] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *J. Comput. Phys.*, vol. 378, pp. 686–707, Feb. 2019.

[36] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Rev. Phys.*, vol. 3, no. 6, pp. 422–440, 2021.

[37] Y. Wen, T. Gao, J. Zhang, Z. Li, and T. Chen, "Encoder-free multiaxis physics-aware fusion network for remote sensing image dehazing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4705915.

[38] B. Li, Q. Hu, M. Gao, T. Liu, C. Zhang, and C. Liu, "Physical informed neural network improving the WRF-CHEM results of air pollution using satellite-based remote sensing data," *Atmos. Environ.*, vol. 311, Oct. 2023, Art. no. 120031.

[39] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10781–10790.

[40] H. Chen and Z. Shi, "Spatial–temporal attention-based method for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1661, 2020.

[41] S. Ji, S. Wei, M. Lu, Z. Shi, Y. Pan, and X. Tang, "A novel supervised change detection method based on deep Siamese convolutional network," *Remote Sens.*, vol. 12, no. 10, p. 1639, 2020.

[42] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[43] A. Kirillov, K. He, R. Girshick, C. Rother, and P. Dollár, "Panoptic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 9404–9413.

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